Malaysia: A Techno-Economic Analysis of Power Generation

April 9, 2025

BloombergNEF

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Section 1. Executive summary

2018

Year when a new utility-scale solar power plant first became cost competitive against a new combined-cycle gas turbine plant in Malaysia

2026

Year when a new solar-with-storage power plant becomes cost competitive against a new combined-cycle gas turbine plant in Malaysia

2.7x

Levelized cost of electricity of a retrofitted combined-cycle gas turbine in Malaysia running on 100% green hydrogen produced in Sarawak compared with solar-with-storage in 2050

Malaysia is aiming to phase out coal power by 2044 and achieve net zero by 2050, all while ensuring energy security and affordability to fulfill soaring power demand and enable economic growth. BloombergNEF's analysis shows that scaling-up renewables is the most economic pathway for Malaysia to achieve these goals. In comparison, retrofitting thermal power plants for hydrogen blending or ammonia co-firing or fitting them with carbon capture and storage technology are both more expensive and less effective in abating emissions.

- This is a pivotal moment for Malaysia to accelerate the decarbonization of its power sector. Efforts will be underpinned by the country's ambitious climate and emission reduction goals outlined in its National Energy Transition Roadmap (NETR) and Nationally Determined Contribution (NDC). Malaysia needs to expand its power system along a path that allows the country to reach its climate targets while maintaining energy security and affordability.
- The NETR calls for restricting the addition of new coal power plants and phasing out existing ones by 2044. However, it also calls for an expansion of the country's gas power plant fleet. The NETR also explores co-firing ammonia or biomass with coal and blending hydrogen with natural gas to reduce emissions. However, **co-firing will not be a cost-effective decarbonization path for Malaysia**, according to BNEF analysis. To achieve tangible emission reductions, a coal power plant would have to be retrofitted to be capable of co-firing biomass or ammonia with coal at energy ratios above 50%. At such high ratios, electricity generation costs will be far higher than renewables. The same applies to retrofitting combined-cycle gas turbines (CCGTs) for hydrogen blending (Figure 1 and Figure 2).
- **Utility-scale solar is already the cheapest source of bulk power generation in Malaysia**. The levelized cost of electricity (LCOE) – the financial measure used by developers and investors to assess the long-term offtake power price needed to recoup project costs and meet the equity investment hurdle rate – for a new utility-scale solar project in Malaysia today ranges from $33 to $61 (152-285 ringgit)[1] per megawatt-hour (MWh) on a real 2024 dollar basis, compared with $78-89/MWh for a CCGT and $56-77/MWh for a coal power plant.
- Solar can be paired with battery storage to address intermittency and provide ancillary services to the grid. **Solar-with-storage will achieve a lower LCOE than new gas and coal power plants by 2026 and 2028, respectively.**
- Malaysia has no plans to install wind power plants. Due to relatively low wind speeds and large variability caused by monsoon seasons, wind power is less viable in Malaysia.
- Malaysia's Sarawak state aims to produce green hydrogen using its abundant hydropower. BNEF's analysis suggests that electrolysis run with hydro-dominated grid power could produce domestic low-carbon hydrogen more cheaply than imports. However, using such cheaper clean hydrogen for reducing emissions from thermal power plants will still not be economical. **Clean hydrogen needs to be prioritized for hard-to-abate sectors where direct electrification is not possible.** Direct renewable energy use is far more effective and affordable to decarbonize the power sector.
- *(May 8, 2025: Corrected Figure 50 chart and related text on page 41. June 12, 2025: Corrected unit for variable operational expenditure on page 30 to $/MWh.)*

[1] Currency conversion on a real 2024 basis assumes $1 = 4.6723 Malaysian ringgit.

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Figure 1: Levelized cost of electricity for new renewables and retrofitted combined-cycle gas turbines for hydrogen in Malaysia, compared against the short-run marginal cost of existing combined-cycle gas turbines

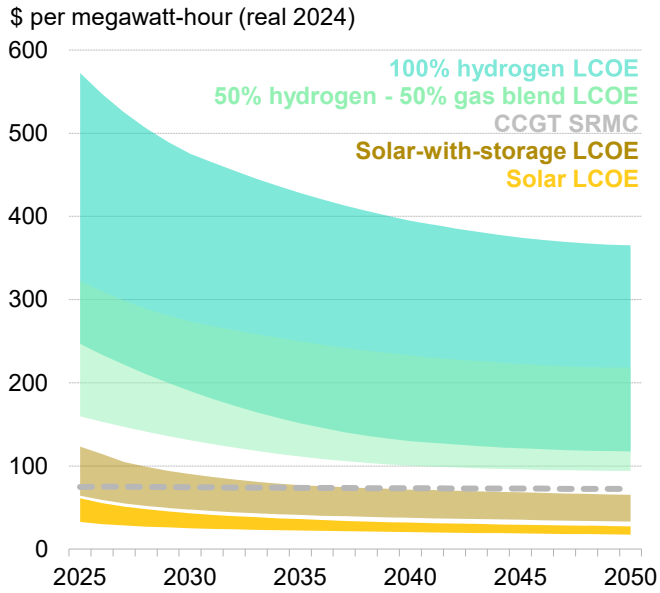
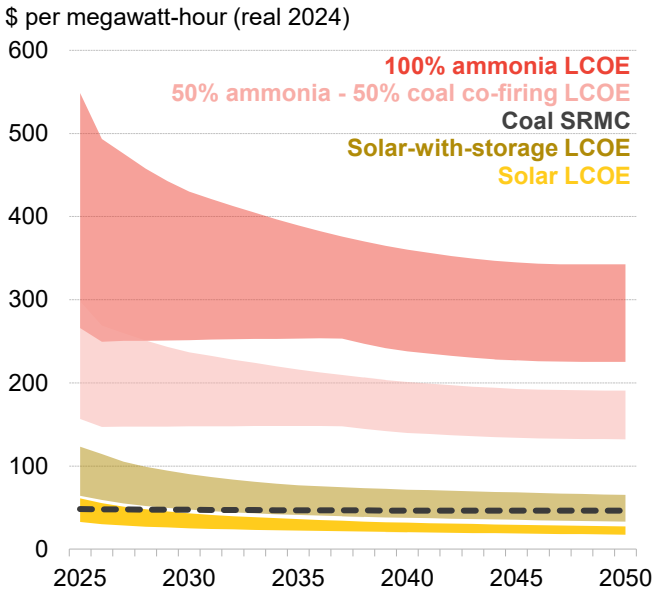


Figure 2: Levelized cost of electricity for new renewables and retrofitted coal power plants for ammonia in Malaysia, compared against the short-run marginal cost of existing coal plants



Source: BloombergNEF. Note: Blending and co-firing ratio is based on energy content. Storage refers to four-hour lithium-ion battery energy storage systems. Inflection points in the levelized costs of electricity (LCOEs) bands for 100% ammonia and co-firing 50% ammonia with 50% coal are years when producing power with domestic ammonia becomes more affordable than with imported molecules. SRMC is short-run marginal cost.

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Section 2. Introduction

Coal dominates Malaysia's power generation mix today

Malaysia has gradually grown its power generation capacity over the last decade to meet rising electricity demand, particularly in the industrial and residential sectors. The country has historically been reliant on coal, gas and hydro power. As of 2022 – the most recent year for which Malaysia has published official data – Malaysia's total **installed capacity** stood at 39.5 gigawatts (GW), 42% of which came from gas-fired power plants, followed by coal (33%) and hydro power facilities (16%) (Figure 3). Despite its lower capacity, coal dominates the power **generation mix** as the country's coal plants are operated at a higher average capacity factor of 74% between 2014 and 2022, compared to 50% for the gas plants (Figure 4).

Malaysia's power system is run by three power utilities: Tenaga Nasional Berhad (TNB) in Peninsular Malaysia, and Sarawak Energy and Sabah Electricity in East Malaysia. Power markets in the Borneo states of Sarawak and Sabah use a single buyer model, with their utilities acting as the sole power procurer, transmitter, distributor and retailer. Sarawak Energy is also the only power producer in its eponymous state, while generation in Sabah is done by both Sabah Electricity and independent power producers (IPPs). Peninsular Malaysia runs a partially liberalized power market. Nearly all the power produced by TNB and IPPs goes through Single Buyer, a ring-fenced department of TNB. However, some commercial and industrial consumers can procure clean power directly from third-party generators starting from October 2024.

Figure 3: Malaysia's historical installed power capacity

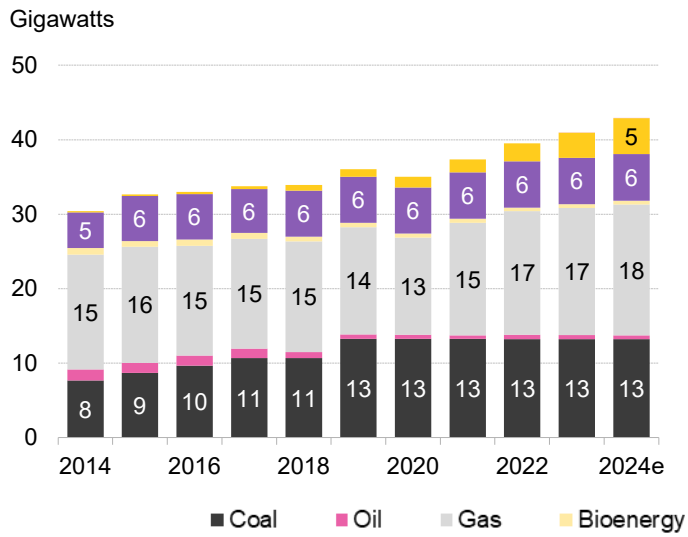
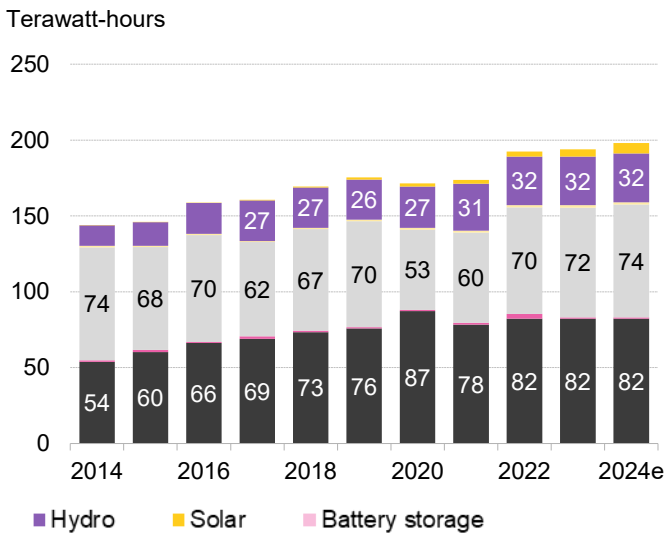


Figure 4: Malaysia's historical electricity generation mix



Source: BloombergNEF, Energy Commission. Note: Figures for 2023-2024 are BNEF estimates.

Malaysia's ambitious climate targets and progress

In September 2021, Malaysia's then-Prime Minister Ismail Sabri Yaakob announced a goal of **achieving net-zero emissions** "by 2050 at the earliest", the most aggressive goal among Southeast Asian countries at the time. Malaysia also joined the Global Methane Pledge to reduce methane emissions by 30% by 2030 from 2020 levels. More recently, the country in 2023 signed the UAE Consensus closed at the United Nations climate summit (COP28) to triple renewables

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and double energy efficiency by 2030. Last year, Malaysia also joined COP29's Global Energy Storage and Grids Pledge to globally deploy 1,500GW of energy storage and add or refurbish 25 million kilometers of grid infrastructure by 2030.

Malaysia has since put in place several strategies to support its net-zero goal, including the **National Energy Transition Roadmap** (NETR) and the **National Climate Change Policy 2.0** (NCCP). In addition, the government is drafting Malaysia's first National Climate Change Bill.

Malaysia aims for a 45% emissions intensity cut from 2005 levels by 2030

In July 2021, Malaysia submitted an update to its first **Nationally Determined Contribution** (NDC), which outlines a country's emissions reduction plan and targets to help the world achieve the goals of the Paris Agreement. It committed to an unconditional target of 45% economy-wide emissions intensity reduction by 2030 compared to 2005 levels. As of March 2025, Malaysia has not submitted its second NDC, which should include its decarbonization target to 2035.

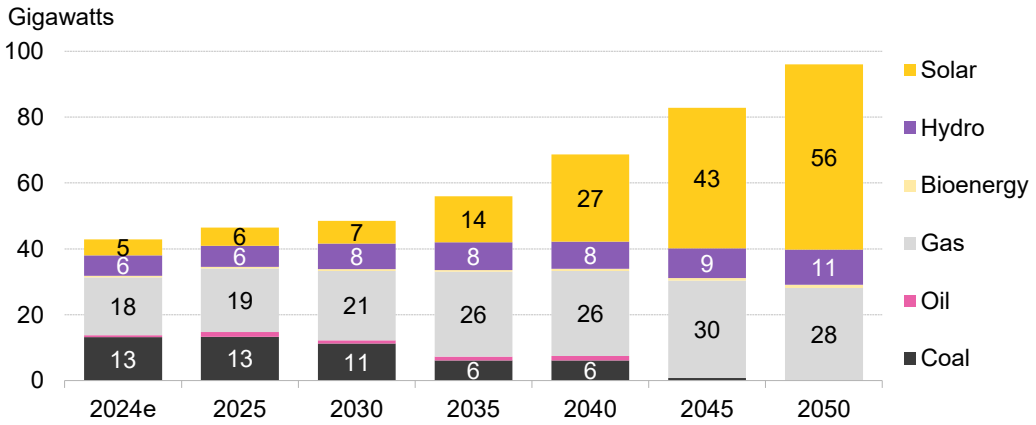
Malaysia's 2025-2050 energy decarbonization pathway

Malaysia's installed solar capacity is projected to grow 17-fold between 2022 and 2050

The 2023 NETR outlines a path to cut the emissions of Malaysia's power generation system to mid-century. It envisions installed capacity to grow by 2.5 times to 97GW in 2050 from 39GW in 2023, with solar and gas leading generation capacity additions (Figure 5). Solar is projected to undergo a 17-fold expansion from 3.3GW in 2022 to 56GW in 2050, including at least 2.5GW of floating solar by 2040. To accommodate this large proportion of electricity generated from solar, the roadmap includes the development of battery energy storage systems as one of its flagship projects. However, Malaysia has not announced a specific target capacity or the timeline for energy storage deployment.

Although the NETR calls for renewables to constitute 70% of installed capacity by 2050, it also expects Malaysia's 28GW gas fleet to contribute the majority of electricity generated due to solar power's lower capacity factor of about 16-18%. Coal capacity is projected to shrink to less than half of current levels by 2035, and then to almost zero by 2045. This is broadly in line with the government's commitment to phase out coal power by 2044. Similarly, diesel-fired plants are also scheduled to vanish by 2045. In contrast, hydropower is expected to nearly double in capacity by 2050 from today. The new hydro plants will most likely be constructed in Sarawak and Sabah, home to 87% of Malaysia's hydropower potential.

Figure 5: Malaysia's estimated installed power generation capacity, historical 2024 and projected 2025-2050 under Malaysia's National Energy Transition Roadmap



Source: BloombergNEF, National Energy Transition Roadmap. Note: Figures for 2024 are BNEF estimates.

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Wind power does not feature in Malaysia's energy transition roadmap

Wind is not considered a viable power source in Malaysia's energy roadmap. Due to its position on the equator, Malaysia has poor wind resources. The country's average wind speed is about 2.7 meters per second at an altitude of 100 meters above sea level, below the 4.5m/s threshold that is generally considered essential to obtain economical wind power. A few areas have sufficiently fast average wind speed, but the high variability introduced by monsoon seasons renders even those areas unappealing for wind power development with current technology and capital expenditure levels.

Beyond expanding renewables capacity, the NETR also calls for the development of carbon capture and storage (CCS) technology, biomass and ammonia co-firing with coal, and hydrogen blending with natural gas to decarbonize thermal power plants.

Malaysia needs to grow its power system in a manner that allows it to attain its climate goals while maintaining energy security and affordability. This report examines the levelized cost of electricity (LCOE) for the different power generation technologies applicable for Malaysia, namely solar, wind, CCGTs and coal power plants. It also looks at the economics of Malaysia's proposed co-firing, fuel blending and CCS strategies for its existing coal and gas power plants. Beyond LCOE, the report also examines the advantages and disadvantages of each technology in terms of impact on Malaysia's energy security and affordability, as well as emissions.

Levelized cost of electricity

LCOE refers to the long-term offtake power price on a per megawatt-hour (MWh) basis required to recoup all project costs to meet the equity investment hurdle rate. BNEF uses its proprietary *Energy Project Asset Valuation Model* (web | terminal) to calculate the LCOE based on input data relevant for each technology in consideration of the location where the project would be built. The calculation is based on a project finance schedule accounting for the full life of the project. This allows us to capture the project cost impact of the timing of cash flows, development and construction costs, multiple stages of financing, interest and tax implications of long-term debt instruments and depreciation, among other factors. For the input parameters used in the LCOE calculations in this report, refer to Appendix A.

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Section 3. Economic analysis

Utility-scale solar already outcompetes new coal and gas plants in Malaysia in terms of economics. The LCOE for solar in the country has been declining sharply since 2019, thanks to the technology's increased deployment in Malaysia and falling solar equipment prices globally.

Meanwhile, the cost of electricity generation for CCGTs is subject to greater volatility as Malaysia gradually becomes more reliant on LNG imports. Using biomass, green hydrogen or ammonia as a co-firing fuel will not be a cost-effective route to decarbonize existing thermal power plants.

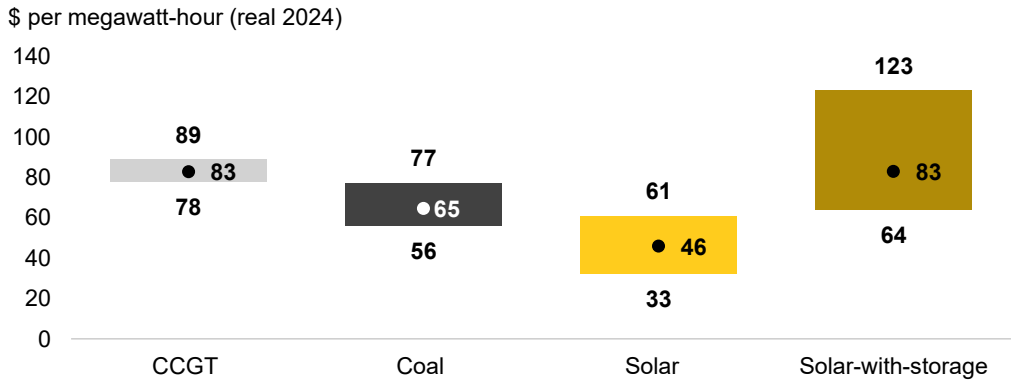
3.1. Solar is the most affordable source of electricity

Utility-scale solar is already the cheapest source of new bulk generation in Malaysia

Decreasing equipment costs and supportive government programs have driven solar power's falling costs

The LCOE for a new utility-scale solar plant currently stands at $33-61/MWh (in real 2024 dollar terms), making it the cheapest source of new bulk electricity generation in Malaysia (Figure 6).[2] The country's Large Scale Solar auctions have been driving large volumes of solar deployment, supported by other programs such as Net Energy Metering, Solar Self-Consumption, the Corporate Green Power Programme and Corporate Renewable Energy Supply Scheme. The increase in activity, declining equipment cost, build-up of domestic solar manufacturing capacity and project development know-how have driven down the LCOE of solar power significantly. The cost advantage of solar over coal and gas will only grow over time (Figure 7 and Figure 8).

Figure 6: Levelized cost of electricity of new power plants in Malaysia in 2025, by technology



Source: BloombergNEF. Note: Solar-with-storage is modeled with a four-hour battery. The low, mid and high points of our solar-with-storage estimate assume a 25%, 50% and 100% capacity ratio between the battery and the solar plant, respectively. CCGT is combined-cycle gas turbine.

[2] Our LCOE estimates are based on real-world project financing data. As there are many more ground-mounted utility-scale solar projects in Peninsular Malaysia compared to in the Eastern Malaysian states of Sarawak and Sabah, the solar and solar-with-storage LCOE data shown in this report mainly reflect projects in Peninsular Malaysia. However, we do not expect a notable difference in the cost of solar power (with and without battery storage) between Peninsular and Eastern Malaysia, as by far the strongest determinants of the LCOE are capacity factor and the costs of equipment and capital, which do not differ significantly across the two regions. LCOE estimates for other power generation technologies apply to all parts of Malaysia.

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Figure 7: Levelized cost of electricity of new power plants in Malaysia in 2030, by technology

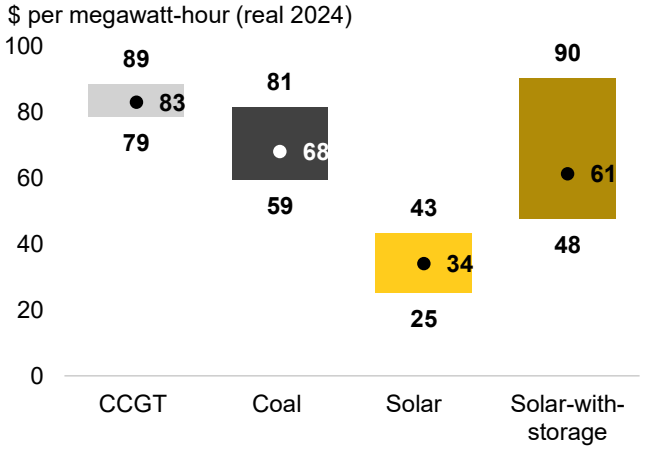
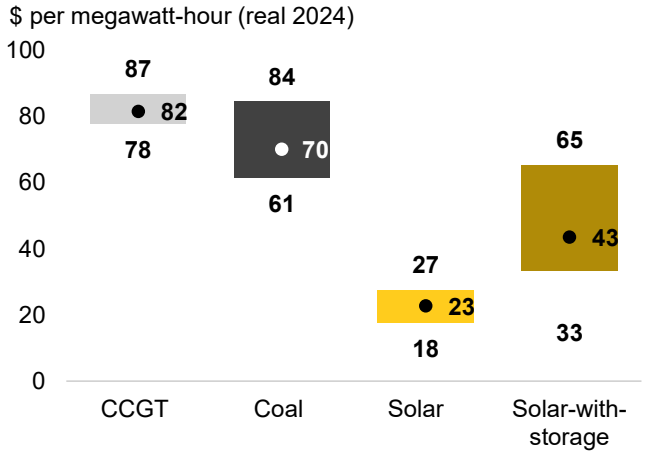


Figure 8: Levelized cost of electricity of new power plants in Malaysia in 2050, by technology



Source: BloombergNEF. Note: Solar-with-storage is modeled with a four-hour battery. The low, mid and high points of our solar-with-storage estimate assume a 25%, 50% and 100% capacity ratio between the battery and the solar plant, respectively. CCGT is combined-cycle gas turbine.

Solar with battery storage will be cheaper than new gas and coal plants by 2026 and 2028

Solar with batteries becomes cheaper than thermal power plants by 2028

Relative to thermal power plants, solar power suffers from a low capacity factor and intermittency. These problems can be solved by pairing solar with battery storage. Despite the additional capital expenditure required for the storage system, BNEF expects a solar plus four-hour battery project to become cost-competitive against a new gas and coal plant by 2026 and 2028 in Malaysia (Figure 9 and Figure 10). The LCOE of such a system is expected to fall to $48-90/MWh by 2030 and $33-65/MWh by 2050, thanks to declining costs of solar modules and lithium-ion batteries.

Figure 9: Levelized cost of electricity of new solar-plus-storage and combined-cycle gas turbine plants in Malaysia

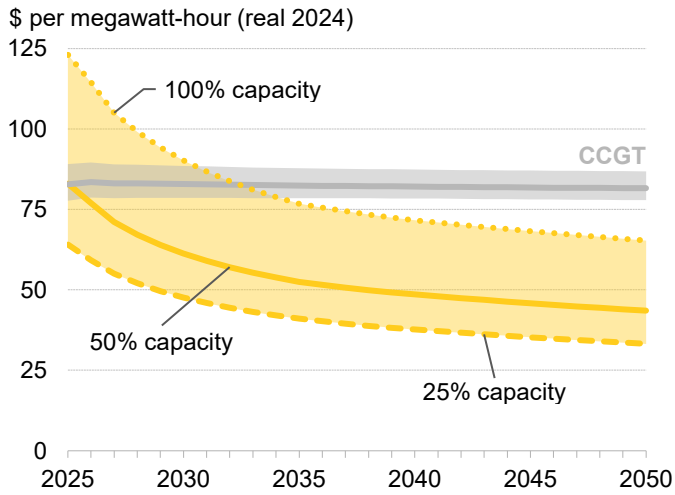
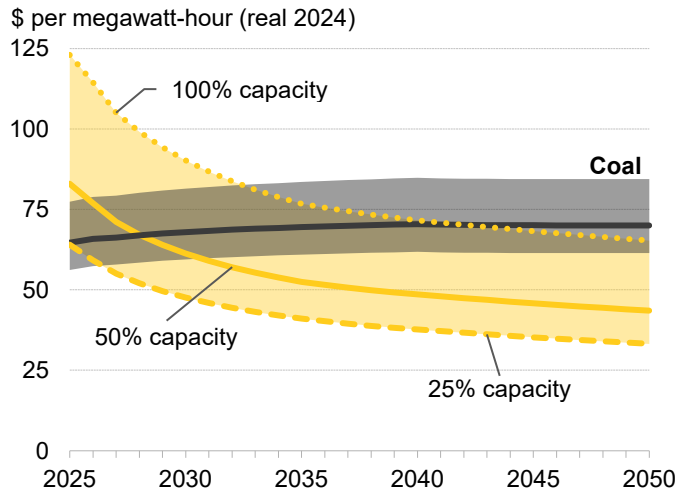


Figure 10: Levelized cost of electricity of new solar-plus-storage and coal power plants in Malaysia



Source: BloombergNEF. Note: Solar-with-storage levelized cost of electricity is modeled with a four-hour battery. The low, mid and high estimates of BNEF's forecast assume a 25%, 50% and 100% capacity ratio between the battery and the solar plant, respectively.

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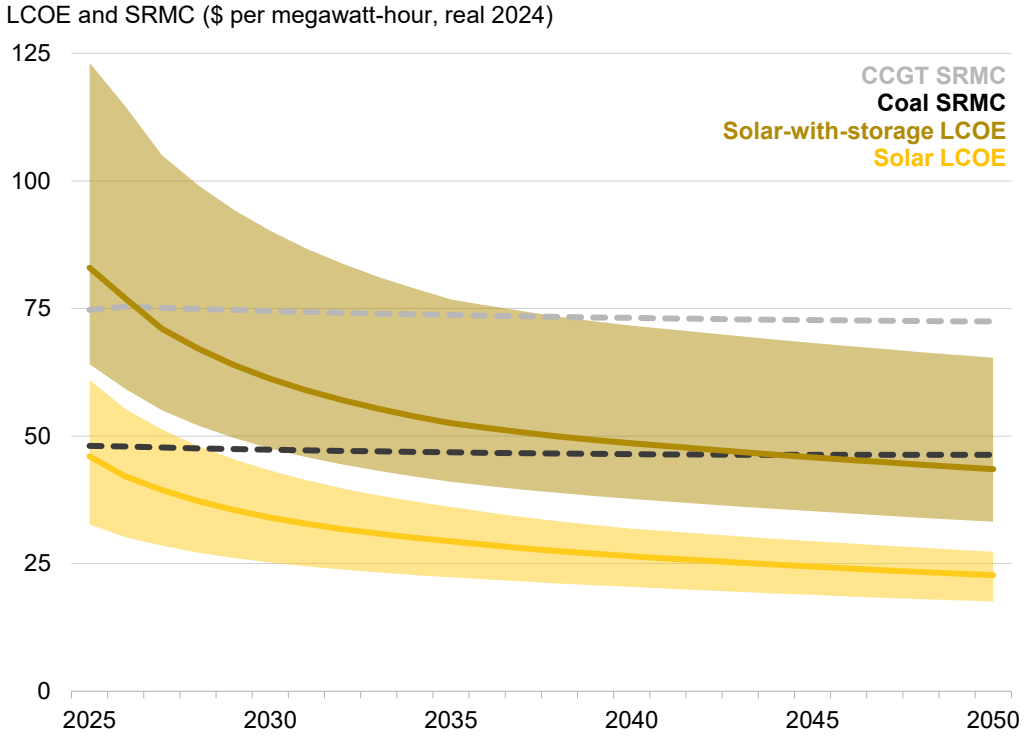
Power from a new solar plant is cheaper than the operational cost of existing gas power plants today, solar-with-storage to follow in 2027

A new solar power plant in Malaysia can already generate electricity more cheaply than existing gas and coal power plants

The 2025 benchmark LCOE of $46/MWh for new ground-mounted solar is lower than the short-run marginal costs (SRMCs) of existing gas ($75/MWh) and coal ($48/MWh) power plants in Malaysia today (Figure 11). Meanwhile, the cost of firmed power from solar-with-storage plants will be cheaper than gas and coal SRMCs by 2027 and 2045, respectively. Coal and gas power generation also face future fuel price volatility. Higher-than-expected fuel costs would further raise the SRMCs of coal and gas plants already in operation, further increasing the economic appeal of renewables.

Figure 11: Levelized cost of electricity of new solar and solar-with-storage plants compared with short-run marginal cost of existing coal and combined-cycle gas turbine plants in Malaysia



Source: BloombergNEF. Note: LCOE is levelized cost of electricity and SRMC is short-run marginal cost. CCGT is combined-cycle gas turbine.

3.2. Retrofitting thermal power plants for hydrogen and ammonia

A few countries, most notably Japan and South Korea, are considering co-firing coal with ammonia and blending natural gas with hydrogen to lower emissions from existing thermal power plants (Figure 12). Malaysia is also exploring this strategy through a partnership between the TNB and its national oil and gas company Petronas, as outlined in the NETR. Molecules such as hydrogen and ammonia do not release carbon dioxide during combustion given the absence of carbon in their molecular chemistry. Still, such approaches entail higher risks (see Section 4) and costs than renewables.

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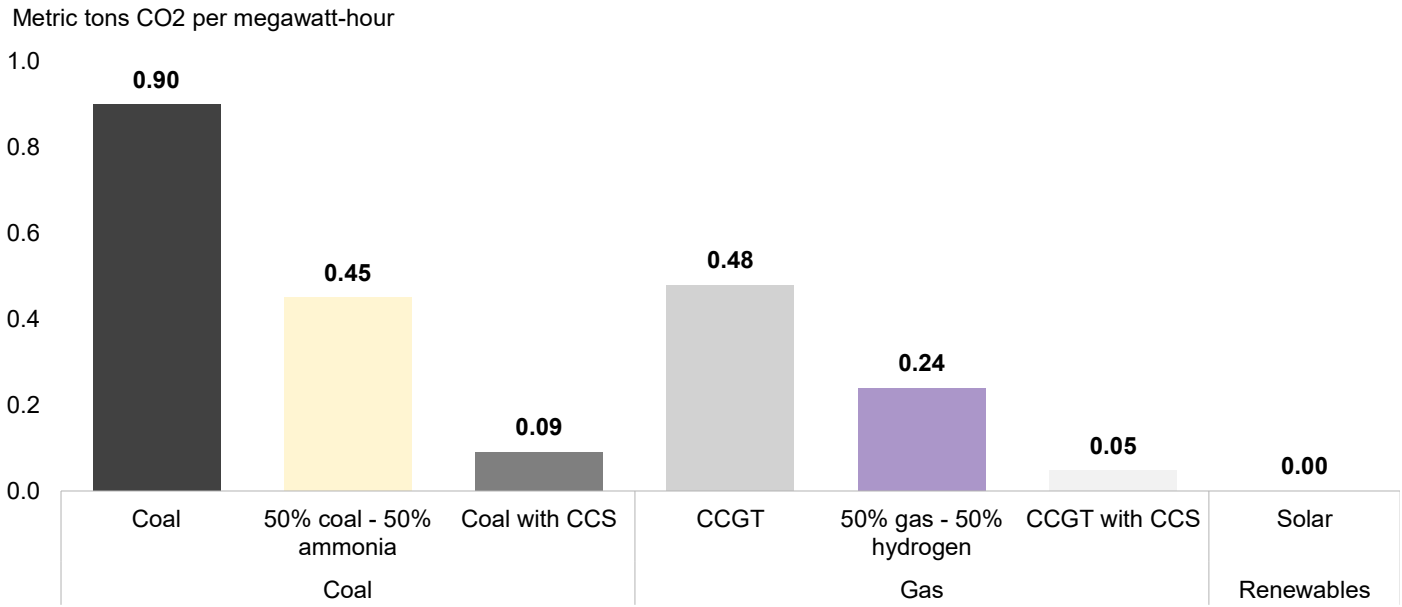
Tests of ammonia co-firing at commercial coal power plants have so far been limited to 20% ammonia on an energy content basis, most recently done by Jera and IHI at the 1GW Hekinan-4 plant in Japan. At such low levels of co-firing, the emissions reduction from the coal power plant is marginal (See Appendix D for more details). The commercial feasibility of ammonia co-firing at ratios higher than 20% is still highly uncertain.

Figure 12: CO2 emission intensity of coal, gas and solar power plants during electricity generation



Source: BloombergNEF. Note: Blending and co-firing ratio is based on energy content. Assumes 90% capture rate for carbon capture and storage (CCS). CCGT is combined-cycle gas turbine.

Environmental benefits of hydrogen blending and ammonia co-firing for power generation are limited

Gray hydrogen production and combustion emit more carbon dioxide than burning gas

To achieve significant carbon dioxide emission reductions from a thermal power plant, the ratio of hydrogen to natural gas as well as ammonia to coal needs to be very high (Figure 13 and Figure 14). Additionally, the hydrogen – and its derivative ammonia – would have to be produced with low emissions. This could either be green hydrogen, which is produced by splitting water to hydrogen and oxygen gases using renewable electricity (electrolysis), or blue hydrogen produced from fossil fuels with nearly all the associated emissions mitigated through carbon capture and storage (CCS). Gray hydrogen produced from fossil fuels without CCS is not a viable option as its production and combustion emit more carbon dioxide than burning natural gas.

Hydrogen blending and ammonia co-firing could also lead to increased risks of air pollution. Since hydrogen and ammonia burn hotter than fossil fuels, the nitrogen and oxygen present in the air during their combustion react at a higher rate, leading to more nitrogen oxide (NOx) emissions. NOx is a class of air pollutants that contributes to the greenhouse gas effect indirectly as well as to rain acidification. These combustion technologies also emit nitrous oxide (N2O), a highly potent greenhouse gas with a global warming potential 273 times greater than that of carbon dioxide over a 100-year timescale.

Retrofitted thermal power plants for hydrogen or ammonia combustion would thus need an investment in technologies to capture NOx and N2O emissions to ensure greenhouse gas emission reduction benefits. This further highlights the poor economics of this strategy.

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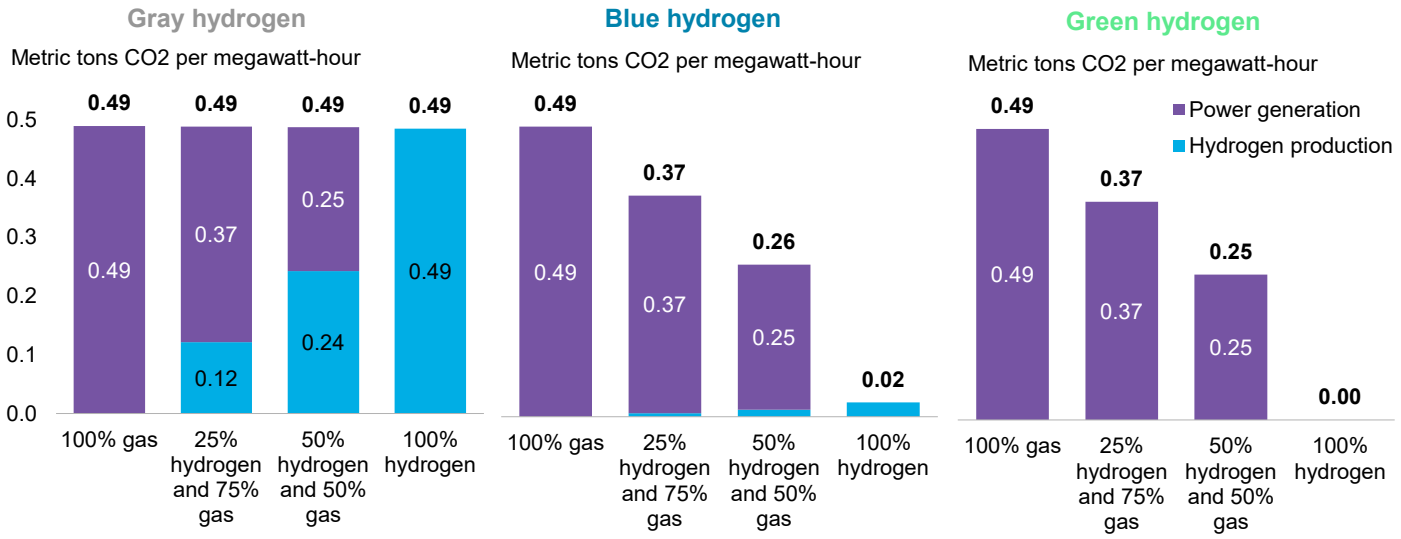
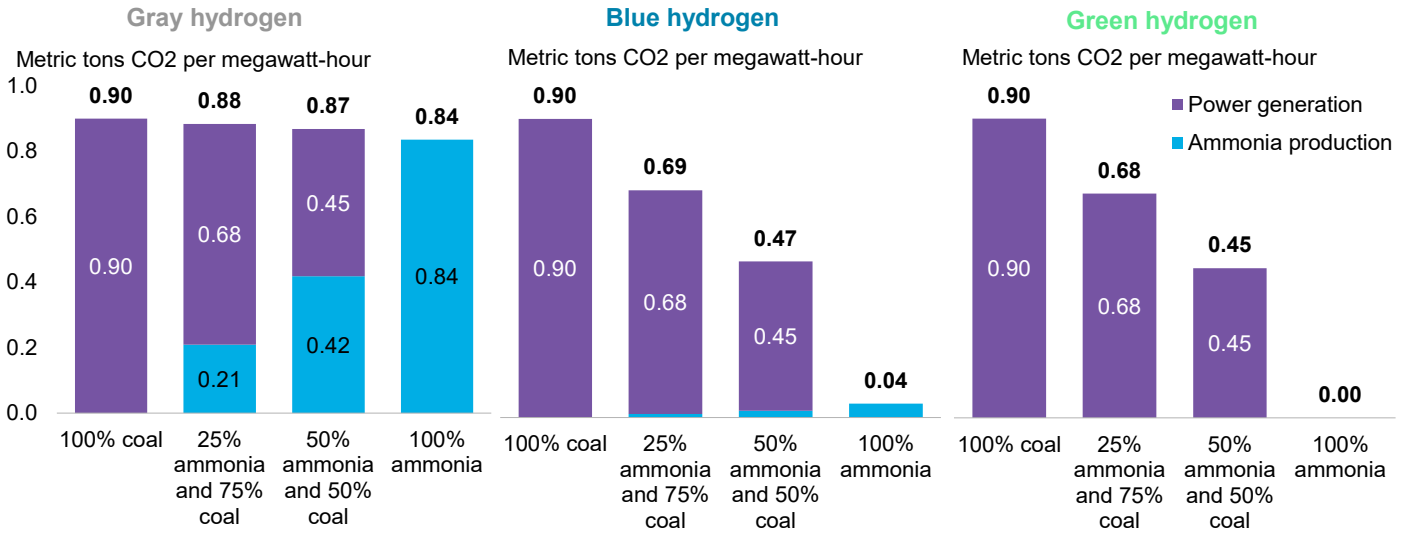
Figure 13: Combined-cycle gas turbine carbon dioxide emissions intensity during electricity generation depending on fuel type



Source: BloombergNEF. Note: Blending ratio is based on energy content.

Figure 14: Coal power plant carbon dioxide emissions intensity during generation depending on fuel type



Source: BloombergNEF. Note: Co-firing ratio is based on energy content.

Renewables are a more economical decarbonization pathway than using hydrogen and ammonia as fuel for thermal power plants

Retrofitting thermal power plants costs more than using solar-with-storage today, even at low co-firing and blending ratios of clean ammonia and hydrogen (Figure 15). Moreover, to achieve significant emissions reduction, thermal power plants must be retrofitted for at least 50% combustion of hydrogen or ammonia, which would be far more expensive than renewables. For the cost parameters used in the adjustments for retrofits of fossil fuel plants, see Appendix A.

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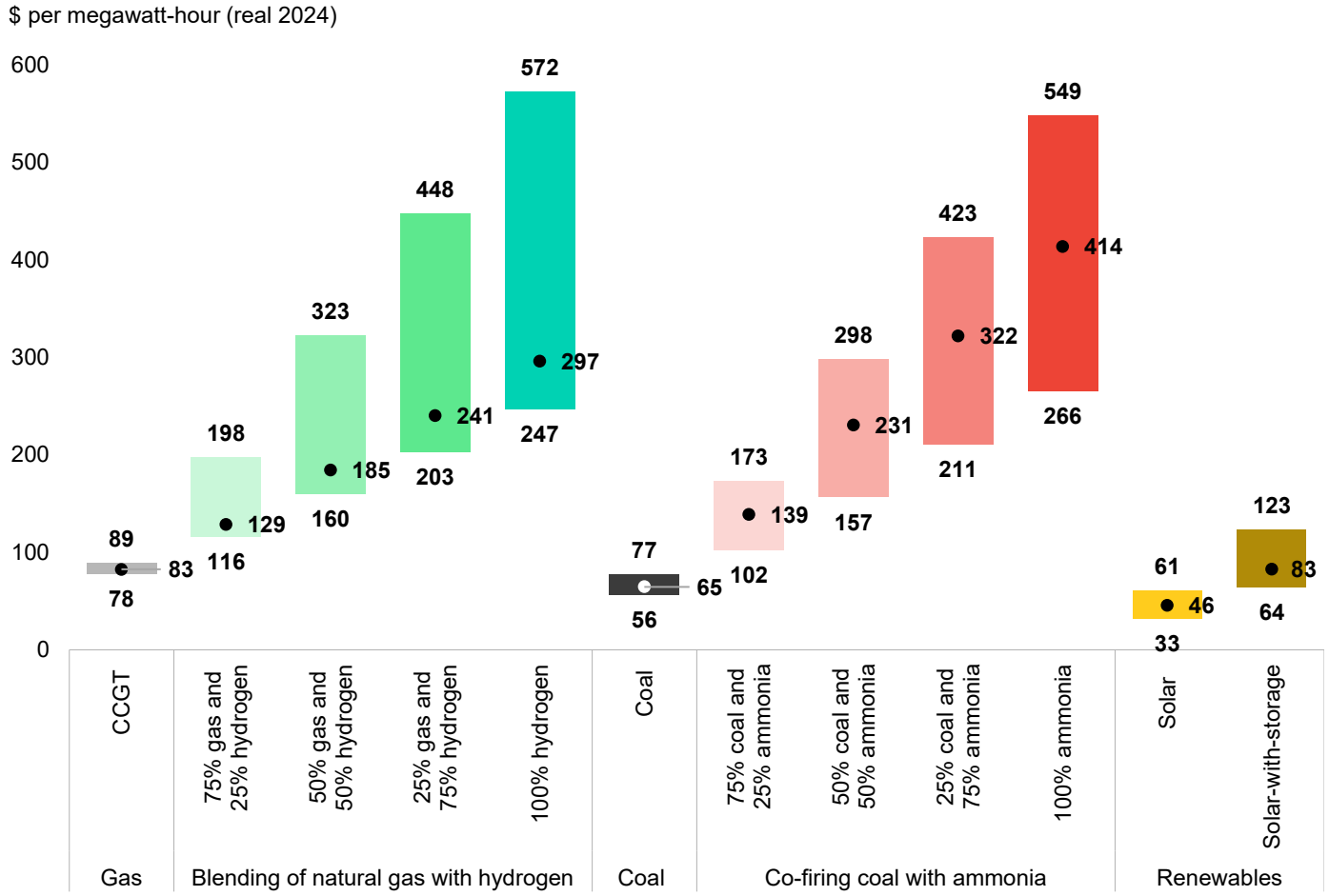
Figure 15: Levelized cost of electricity for coal and gas plants retrofitted for cleaner fuels, compared with new solar-with-storage power plant in Malaysia in 2025



Source: BloombergNEF. Note: Blending and co-firing ratio is based on energy content. CCGT is combined-cycle gas turbine.

Developing solar plants and battery systems produces power more cheaply than co-firing low-carbon hydrogen or ammonia

By the end of this decade, solar with batteries would be the cheapest dispatchable power generator (Figure 16). The economic competitiveness of solar paired with batteries compared with hydrogen or ammonia combustion improves further over the following years (Figure 17). Although Malaysia can produce low-carbon hydrogen at relatively low costs compared to other countries, BNEF analysis suggests the produced hydrogen should not be used for power generation. Rather, **hydrogen is more suited for applications where direct electrification is unfeasible or challenging**, such as heavy-duty trucking, shipping, high-heat industrial processes and production of iron, steel, fertilizers, chemicals and sustainable aviation fuels.

BNEF's analysis assumes that hydrogen production in Malaysia is done through electrolysis in Sarawak, where about 70% of grid power is generated by hydroelectric dams. While this means the electrolysis process is not completely emissions-free, the products are referred to as green hydrogen and ammonia in this report for simplicity. Achieving truly emissions-free hydrogen production would require procurement of renewable electricity through power purchase agreements (PPAs), renewable energy certificate purchases or other methods. For more details on hydrogen and ammonia production that are relevant to Malaysia, see Appendix B (delivered costs of hydrogen and ammonia), Appendix C (production costs of hydrogen and ammonia) and Appendix D (blended fuel prices).

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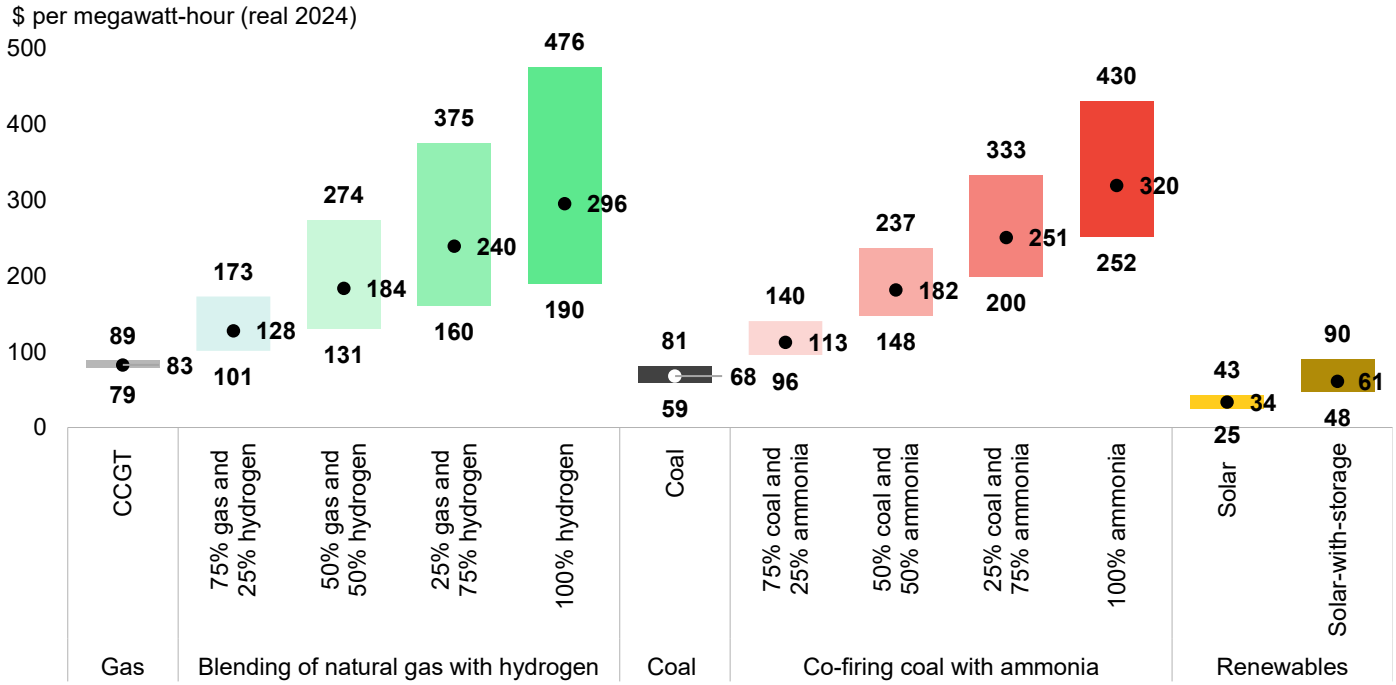
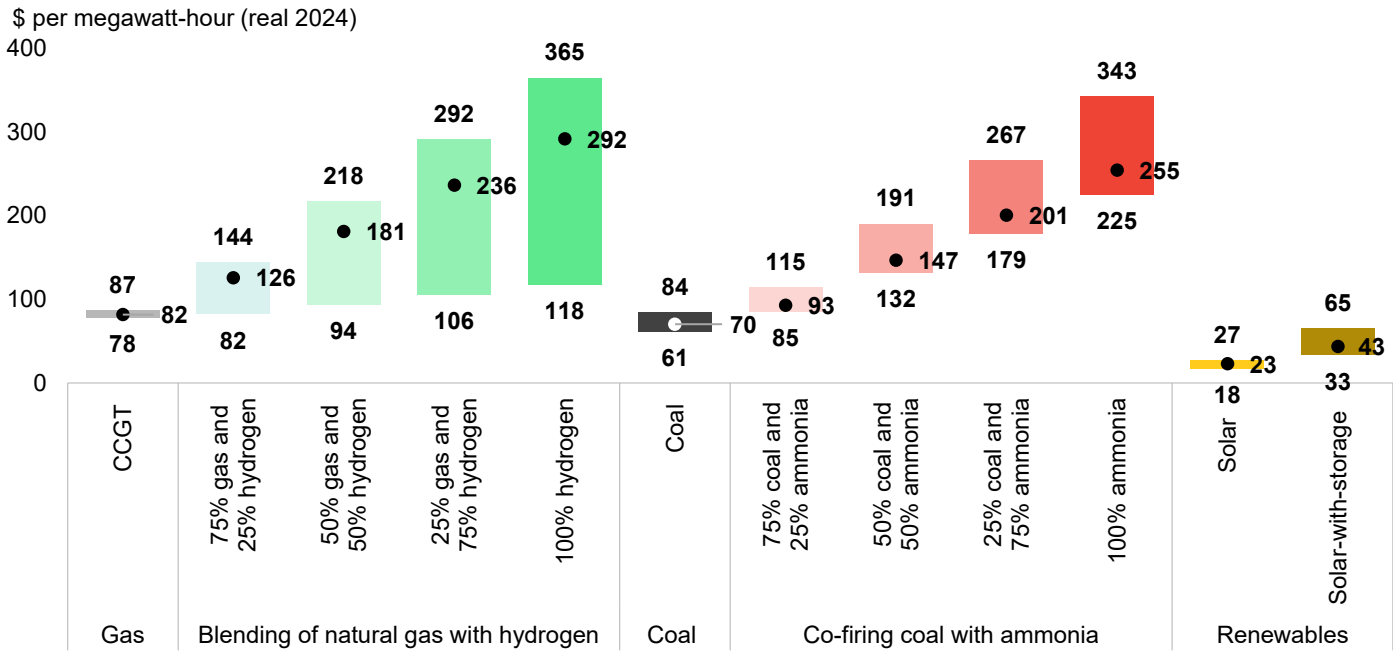
Figure 16: Levelized cost of electricity for coal and gas plants retrofitted for cleaner fuels, compared with new solar-with-storage power plant in Malaysia in 2030



Source: BloombergNEF. Note: Blending and co-firing ratio is based on energy content. CCGT is combined-cycle gas turbine.

Figure 17: Levelized cost of electricity for coal and gas plants retrofitted for cleaner fuels, compared with new solar-with-storage power plant in Malaysia in 2050



Source: BloombergNEF. Note: Blending and co-firing ratio is based on energy content. CCGT is combined-cycle gas turbine.

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3.3. Retrofitting coal power plants for biomass co-firing

Co-firing of biomass with coal entails similar challenges to that of ammonia. At low co-firing ratios, emission reduction benefits are limited. At high co-firing ratios, significant investments and upgrades to the coal power plant would need to be made. In addition, revising existing power purchase agreements to enable co-firing could also be a challenge. In December 2024, Malaysia's Minister of Economy informed the Dewan Negara (Senate) that the NETR's target of achieving 15% biomass co-firing by 2027 needs to be postponed due to the lack of a clause governing co-firing in existing PPAs of thermal power plants.

Securing a sufficient and continuous supply of affordable and sustainable biomass could be challenging

Moreover, securing a sufficient and continuous supply of biomass for high co-firing ratios would also be challenging given the significant increase in feedstock supply required. The viability of using biomass as an alternative fuel supply also hinges on the feedstock price, which is highly sensitive to transportation distance. Despite Malaysia's large agriculture sector, biomass power plants in the country have faced issues with procuring sufficient and reliable feedstock, resulting in small plant sizes, low capacity factors, and relatively high LCOEs.

If there is insufficient biomass feedstock locally, Malaysia will have to turn to imports but will likely find itself facing stiff competition from other countries that are pursuing a similar strategy, such as Japan, South Korea and Indonesia. The latter's state-owned utility Perusahaan Listrik Negara co-fired 1.65 million metric tons of biomass in its coal power plants in 2024 and is projecting a near-doubling to 3 million tons this year to support the country's co-firing ambitions. This may limit the amount of biomass available for imports in the region.

Similar to other countries, Malaysia will need to prioritize its limited supply of bioenergy feedstocks to decarbonize hard-to-abate sectors that have more limited emissions abatement methods, such as by producing clean fuels for aviation and shipping.

3.4. Using carbon capture and storage

Retrofitting existing plants with CCS won't reduce the reliance on fossil fuels and is costlier

Many countries are also considering the use of CCS to reduce emissions from thermal power plants, but there are few operational projects to date. BNEF has modeled the LCOE for several CCS scenarios: greenfield thermal power plants with CCS, as well as retrofitting existing thermal power plants with CCS at 10 years and 15 years after the commercial operation date.

BNEF analysis suggests that a greenfield coal or gas plant equipped with CCS upfront is the most economical scenario (Figure 18 and Figure 19), due to the higher capital expenditure required for retrofits compared to a greenfield project designed with CCS in mind. Retrofitting a thermal power plant 15 years after its commissioning would be slightly cheaper than retrofitting 10 years later, due to the expected reductions in CCS costs. However, this would pose risks for emissions reduction, as the thermal power plant would continue operating with its emissions unabated for an additional five years.

All CCS scenarios in BNEF's analysis are far more expensive than solar-with-storage power in Malaysia. Although CCS scenarios appear to be more economical than co-firing ammonia with coal or blending hydrogen with natural gas at high energy ratios, the amount of potential carbon storage capacity available in Malaysia as well as the feasibility of transporting captured emissions from existing thermal power plants to the carbon storage sites also need to be evaluated. Actual CCS costs in Malaysia could be significantly higher than BNEF's current estimates due to a large uncertainty in the cost of offshore emission storage.

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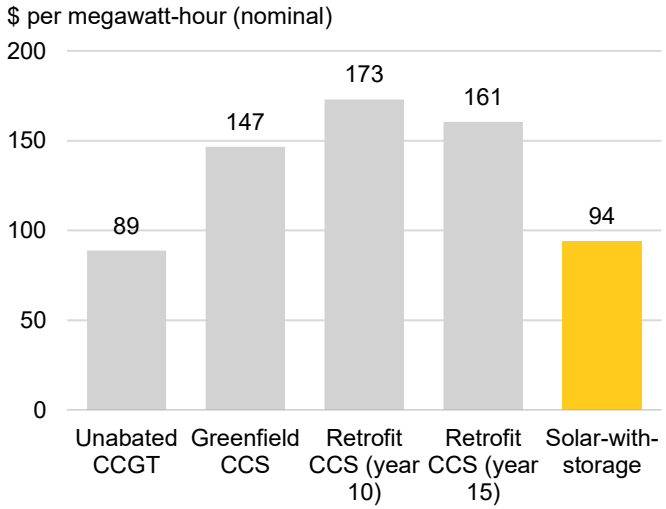
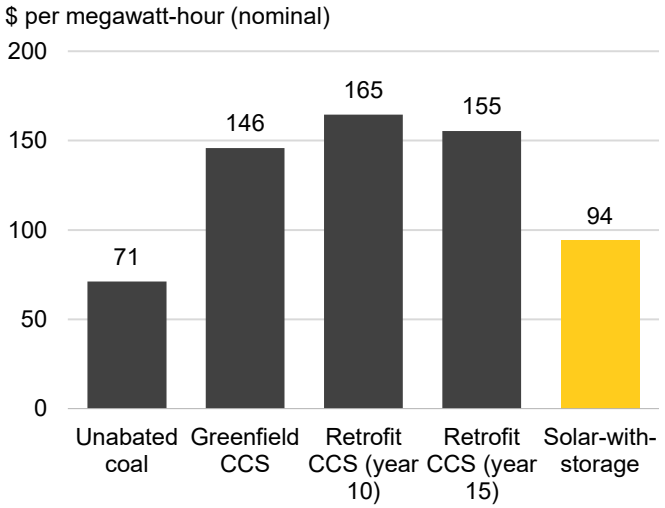
Figure 18: Levelized cost of electricity for new combined-cycle gas turbine plants with and without carbon capture and offshore storage, compared against a new solar-with-storage plant in Malaysia



Source: BloombergNEF. Note: Greenfield CCGT plants are financed in 2024 and expected to commission in 2026. Retrofits at year 10 and 15 refer to 2036 and 2041, respectively. Solar-with storage plant is financed in 2024 and expected to commission in 2025.

Figure 19: Levelized cost of electricity for a new coal plant with and without carbon capture and offshore storage, compared against a new solar-with-storage plant in Malaysia



Source: BloombergNEF. Note: Greenfield coal plants are financed in 2024 and expected to commission in 2028. Retrofits at year 10 and 15 refer to 2038 and 2043, respectively. Solar-with storage plant is financed in 2024 and expected to commission in 2025.

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Section 4. Challenges with using hydrogen as a fuel for electricity generation

The previous section explores the economics of blending low-carbon hydrogen with gas and co-firing ammonia with coal. Here we examine the safety as well as energy security and affordability challenges associated with retrofitting thermal power plants for clean fuels.

4.1. Reliance on hydrogen as a fuel for electricity would increase Malaysia's financial burden

Hydrogen and ammonia are more expensive fuels than gas and coal on an energy-equivalent basis due to these molecules' lower volumetric energy densities. This explains the rise in LCOE at higher ratios of hydrogen or ammonia. Reliance on such fuels could increase power prices and potentially the financial burden on taxpayers if the government or power utility decides to pay the higher costs of these clean fuels by raising regulated electricity tariffs or taxes.

The fuel cost of power generation with 100% local hydrogen could be five times more expensive than natural gas in 2030 and twice as costly in 2050

BNEF estimates a retrofitted 1GW gas power plant running on 50% capacity factor and using 100% hydrogen fuel would need 236,300 tons of hydrogen per year. To source this much hydrogen locally, the annual hydrogen procurement costs per GW would be $1.22 billion in 2030, $0.70 billion in 2040, and $0.49 billion in 2050 (Figure 20 – Figure 22, in real 2024 dollar terms). These would be lower than the cost of imported hydrogen procurement: $1.19 billion-$2.76 billion in 2030, $1.22 billion-$1.95 billion in 2040, and $1.23 billion-$1.63 billion in 2050. Generating the same amount of electricity with natural gas would only take about $0.21 billion-$0.23 billion per GW annually on gas procurement from 2030 to 2050. Fuel procurement for 100% local hydrogen firing would cost about five and two times as much as gas procurement in 2030 and 2050, respectively, potentially leading to higher power tariffs. Imported hydrogen would be costlier still.

Figure 20: Malaysia's annual hydrogen procurement cost per gigawatt of combined-cycle gas turbine power plants in 2030

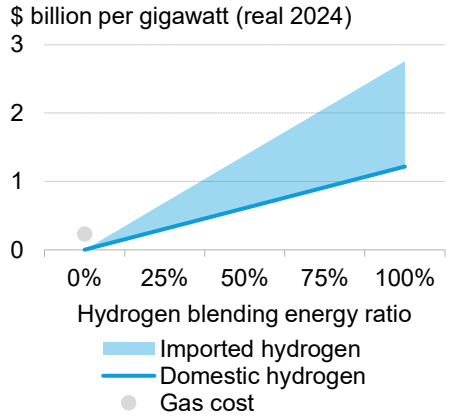


Figure 21: Malaysia's annual hydrogen procurement cost per gigawatt of combined-cycle gas turbine power plants in 2040

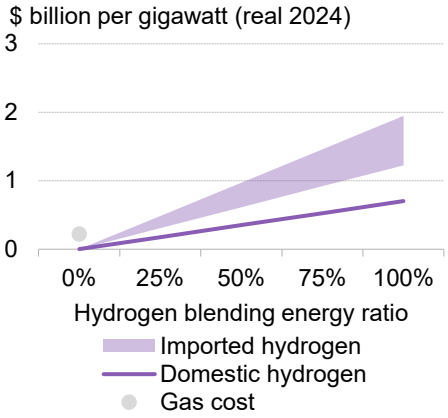
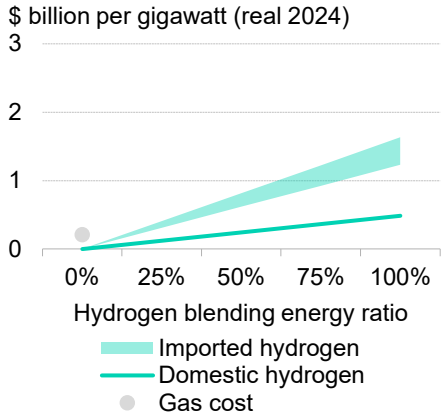


Figure 22: Malaysia's annual hydrogen procurement cost per gigawatt of combined-cycle gas turbine power plants in 2050



Source: BloombergNEF. Note: BNEF estimates 54 kilograms of hydrogen are needed to generate 1 megawatt-hour of electricity. Assumes a CCGT power plant operates at a capacity factor of 50%.

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Ammonia procurement would be at least eight and nine times more expensive than coal procurement in 2030 and 2050, respectively

A retrofitted 1GW coal power plant running on 70% capacity factor in Malaysia would require 1.31 million tons of ammonia for 50% co-firing and 2.63 million tons for 100% ammonia firing. BNEF estimates burning only ammonia in such power plants would require $1.46 billion to $3.64 billion per GW in 2030, $1.50 billion to $2.51 billion in 2040, and $1.34 billion to $2.07 billion in 2050 (Figure 23-Figure 25). On the other hand, burning only coal at a 1GW coal power plant in Malaysia would annually cost $0.18 billion in 2030, $0.17 billion in 2040, and $0.15 billion in 2050 for coal procurement. Ammonia procurement from the cheapest source each year would be eight times more expensive than coal procurement in 2030 and about nine times costlier by 2050.

Figure 23: Malaysia's annual ammonia procurement cost per gigawatt of coal power plants in 2030

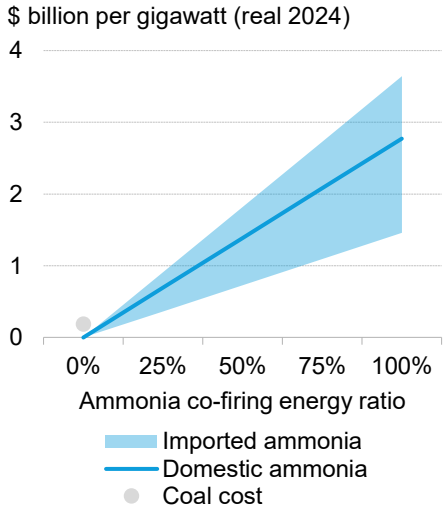


Figure 24: Malaysia's annual ammonia procurement cost per gigawatt of coal power plants in 2040

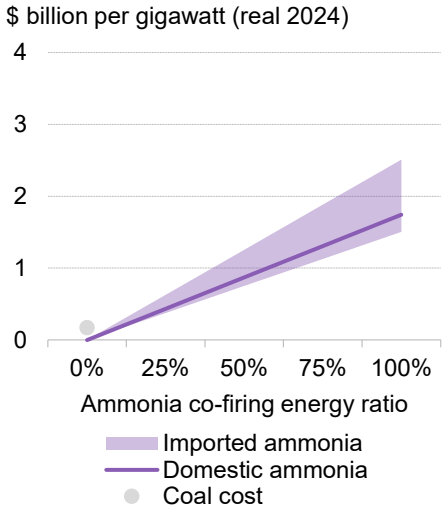
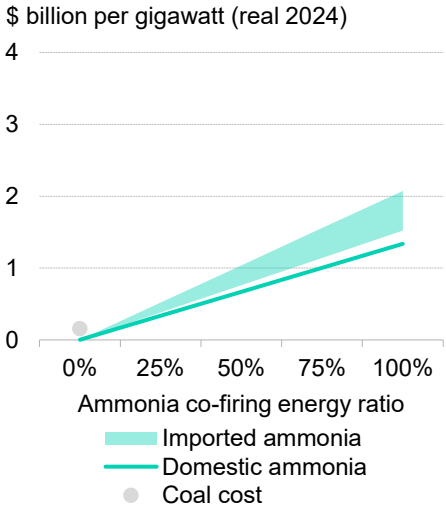


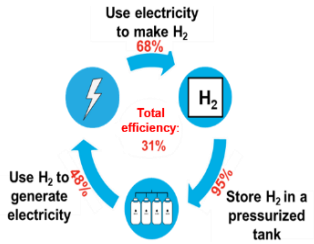
Figure 25: Malaysia's annual ammonia procurement cost per gigawatt of coal power plants in 2050



Source: BloombergNEF. Note: BNEF estimates 0.43 metric tons of ammonia are needed to generate 1 megawatt-hour of electricity. Assumes a coal power plant operates at a capacity factor of 70%.

4.2. Marginal emissions abatement cost for thermal power plants retrofitted for hydrogen and ammonia

Figure 26: Efficiency of hydrogen-to-power cycle



Source: BloombergNEF

Power generation with a blend of domestic green hydrogen and natural gas would be dependent on using locally available renewable electricity to produce hydrogen and then using the hydrogen to produce electricity, a wasteful cycle where more than two-thirds of the energy input is lost (Figure 26). Such indirect use of renewable electricity would be less efficient and much more expensive than just directly using the electricity generated by renewables.

BNEF estimates the marginal emissions abatement cost for 25% domestic hydrogen blending and ammonia co-firing in 2030 to be $220 and $291 per metric ton of carbon dioxide, respectively (Figure 27 and Figure 28). For 50% domestic ammonia co-firing, the abatement cost would be $208/tCO2 in 2040 and $191/tCO2 in 2050. These levies would be a huge financial burden for power plant owners and electricity end-users. If Malaysia were to put in carbon prices anywhere near these levels, power plant owners would likely opt to shut down existing thermal power plants and build cheaper renewables.

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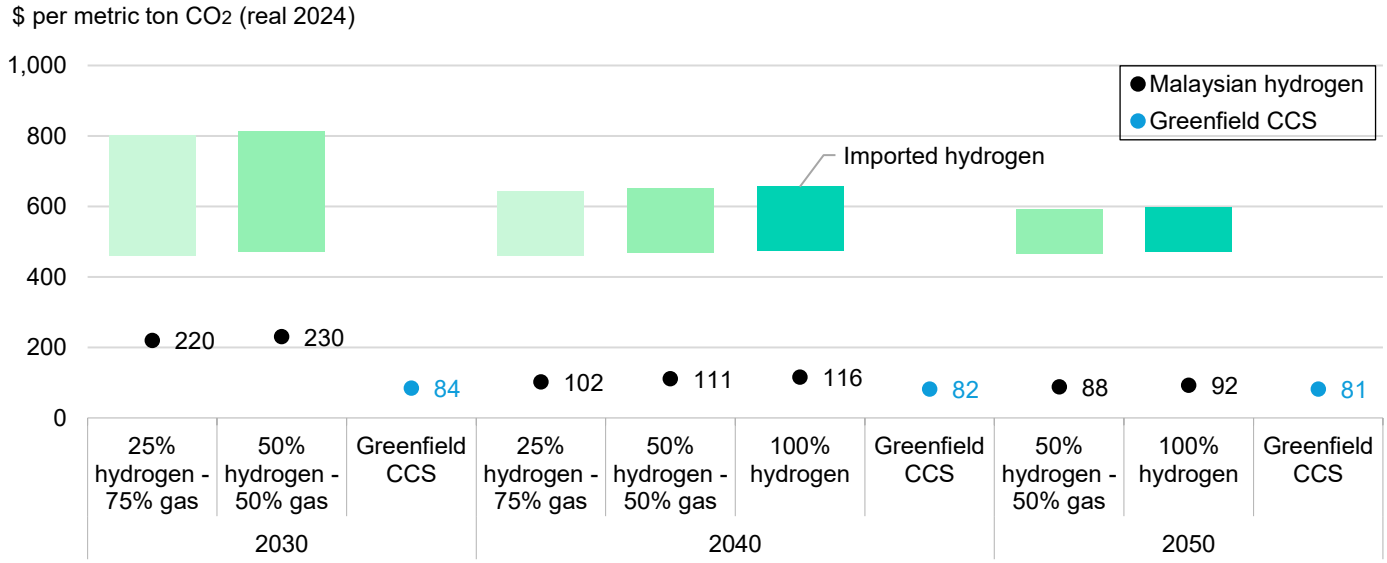
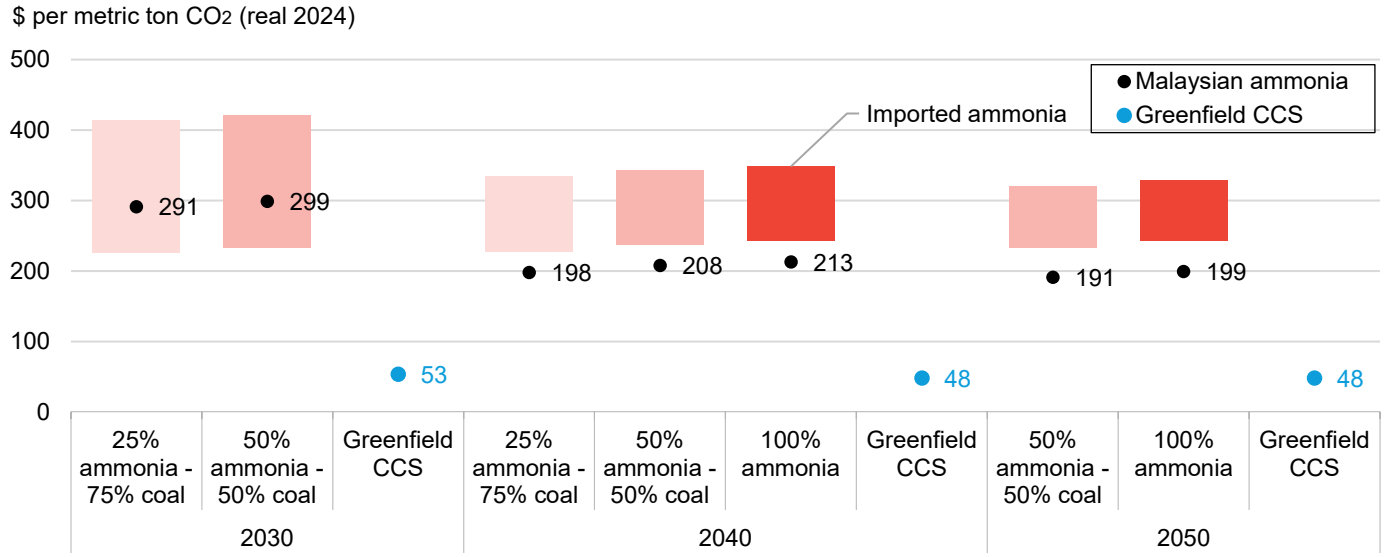
The abatement cost of blending domestic hydrogen and natural gas is relatively affordable compared to ammonia co-firing due to the lower cost premium of domestic hydrogen over gas compared to ammonia over coal. However, hydrogen blending remains less competitive than solar or solar-with-storage, both of which could generate carbon-free power at a lower cost than thermal power plants.

Figure 27: Marginal emissions abatement cost for combined-cycle gas turbines retrofitted for hydrogen blending and fitted with carbon capture and storage technology in Malaysia



Source: BloombergNEF. Note: Blending ratio is based on energy content.

Figure 28: Marginal emissions abatement cost for coal power plant retrofitted for ammonia co-firing and fitted with carbon capture and storage technology in Malaysia



Source: BloombergNEF. Note: Co-firing ratio is based on energy content.

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The marginal emissions abatement costs of greenfield CCS – fitting CCS systems onto coal and gas power plants from the start of their lifetime – are generally lower than those of hydrogen blending and ammonia co-firing in this decade. This remains true until 2050 for ammonia co-firing, while the marginal abatement cost hydrogen blending becomes more competitive with CCS over time as the cost of hydrogen falls.

The abatement cost of CCS is significantly lower for coal plants than for CCGTs as coal plants emit more carbon dioxide, about 90% of which is then captured by the CCS system. However, renewables, such as utility-scale solar and solar-with-storage, remain more economical and effective than adding CCS to new coal or gas plants at cutting power generation emissions. As noted in Section 3.4, actual CCS costs in Malaysia may be significantly higher than BNEF's current estimates, which in turn would result in much higher marginal emissions abatement costs.

4.3. Low-carbon hydrogen and ammonia should be prioritized for hard-to-abate sectors

Green hydrogen should be prioritized for use in hard-to-abate sectors, such as high-heat industrial processes, shipping and aviation

Given the unfavorable economics of using hydrogen and ammonia for power generation in comparison to renewables, **Malaysia needs to prioritize using its low-carbon molecules in applications where electrification is less practical**. In principle, wherever electrification is possible, it will be simpler and more affordable. This includes applications like rail transport, passenger road transport, petrochemicals and cement production.

However, switching from gray hydrogen to low-carbon hydrogen is vital in decarbonizing incumbent uses, such as producing ammonia for the fertilizer and chemical industries, methanol for plastics and chemicals, steel production and oil refining. Hydrogen is also an important vector in the decarbonization of commercial transport through direct and indirect use, such as heavy-duty goods transport, aviation, shipping and production of sustainable fuels.

In shipping, the high energy density of clean hydrogen-derived fuels such as methanol compared to batteries and their ability to be produced at scale make them a key solution to decarbonize maritime trade. Similarly, in aviation, the fuel's high energy-to-mass ratio makes aircraft fueled by pure hydrogen a suitable alternative for medium-distance travel, while aircraft fueled by sustainable aviation fuel made with clean hydrogen can be used for longer distances. Electric airplanes, which can help decarbonize commuter routes, face challenges due to battery size and weight.

4.4. Safety

Ammonia and hydrogen need to be handled with great care due to their high flammability and explosiveness when exposed to heat (Table 1). A leak of liquid ammonia at a poultry plant in China's Jilin province in 2013 caused a fire and resulted in 120 fatalities. In the same year, another ammonia leak killed 15 and injured 25 at a frozen seafood plant in Shanghai, China. In 2017, a hydrogen leak from a coolant at a coal-fired power plant in Ohio, US, caused an explosion that killed one person and injured 10 people. Since hydrogen does not have a distinct odor and color, it is difficult to detect leaks.

These lethal leaks from ammonia and hydrogen have continued to happen in recent years, with accidents being reported in India, the US and the Philippines. In 2021 and 2023, two ammonia

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leaks from an ice plant and a cold storage facility in the Philippines' National Capital Region left a total of three people dead and 113 people hospitalized.

In addition, ammonia must be transported and stored carefully as it could pose a major threat to human health. The molecule reacts with water to form ammonium hydroxide, which is corrosive and damages cells in the body on contact. It is easier to detect an ammonia leak due to its strong odor, but contact with ammonia could be fatal. See *Ammonia's Promise Carries Deadly Dark Side Risks* (web | terminal) for more details.

Table 1: Safety comparison of ammonia, hydrogen and natural gas

	Ammonia	Hydrogen	Natural gas (methane)
Flammability	Flammable	Flammable	Flammable
Explosiveness	May explode if heated	May explode if heated	May explode if heated
Toxicity	Acute poisoning from inhaling, skin/eye/respiratory damages	None; still, high levels of hydrogen could cause a lack of oxygen in bodies	None; still, high levels of methane could cause a lack of oxygen in bodies
Odor	Strong (easy to detect)	None (hard to detect)	None (hard to detect); gas companies add artificial smell
Visibility (color)	Colorless (hard to detect)	Colorless (hard to detect)	Colorless (hard to detect)

Source: BloombergNEF, Globally Harmonized System of Classification and Labeling of Chemical (GHS) classification. Note: Red represents danger. Green indicates no harm.

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Section 5. The way forward for Malaysia

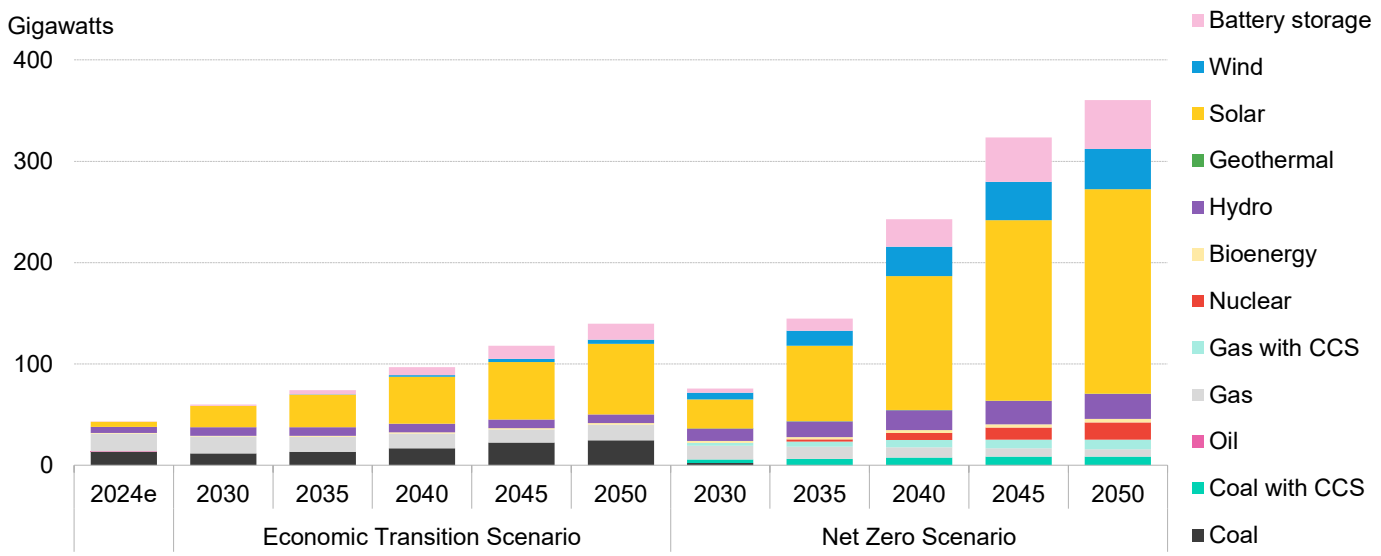
Malaysia's energy transition can be achieved by building renewables, expanding the power grid and limiting new capacity of thermal power plants

Solar power is already an economically preferable option to meet Malaysia's growing power demand while keeping the country aligned with its 2050 net-zero target. Indeed, the country has made good progress in boosting the role of solar in its power generation mix. This has been driven not only by the increasingly favorable economics of solar energy, but also by supportive government programs. In contrast, **expanding Malaysia's thermal power fleet risks the country's energy security and affordability**. Retrofitting thermal power plants for the combustion of hydrogen or ammonia is unlikely to become an economically viable option.

BNEF has modeled two scenarios for Malaysia's future power generation system up to 2050. The **Economic Transition Scenario** (ETS) reflects a world where policymakers pursue an energy transition relying only on historical efficiency trends and economically competitive, commercially at-scale clean energy technologies. In the ETS, Malaysia's power generation capacity grows more than threefold from 2024 to 2050. Solar provides two-thirds of the added capacity due to its advantageous economics. However, as solar power's capacity factor is relatively low, it supplies only 27% of the country's electricity demand at mid-century in the ETS, while coal provides just above 50%. If Malaysia is to reach its goal of completely phasing out coal power by 2044, it would need to rapidly expand its renewables capacity to replace coal generation.

Our **Net Zero Scenario** (NZS) models a cost-effective pathway for Malaysia's power system to reach net zero by 2050. In the NZS, Malaysia's power plant capacity and electricity generation multiply by a factor of 8.6 and 3.6, respectively, between 2024 and 2050. The new power demand is met by a wider range of generation technologies, including renewables, nuclear and thermal plants fitted with CCS. Our modeling shows that the window for achieving net zero by 2050 is rapidly closing. To achieve this goal, the government needs to significantly accelerate clean power deployment efforts, starting immediately (Figure 29 and Figure 30).

Figure 29: Malaysia's installed power generation capacity, estimated 2024 and in BNEF's Economic Transition Scenario and Net Zero Scenario (2030-2050)



Source: BloombergNEF. Note: Figures for 2024 are estimated. CCS is carbon capture and storage.

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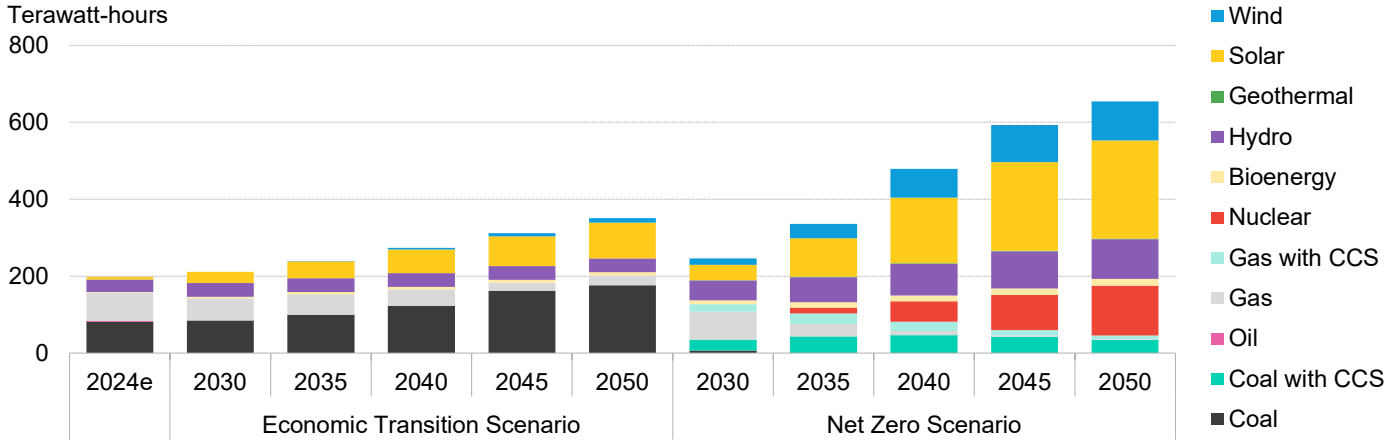
Figure 30: Malaysia's electricity generation mix, estimated 2024 and in BNEF's Economic Transition Scenario and Net Zero Scenario (2030-2050)

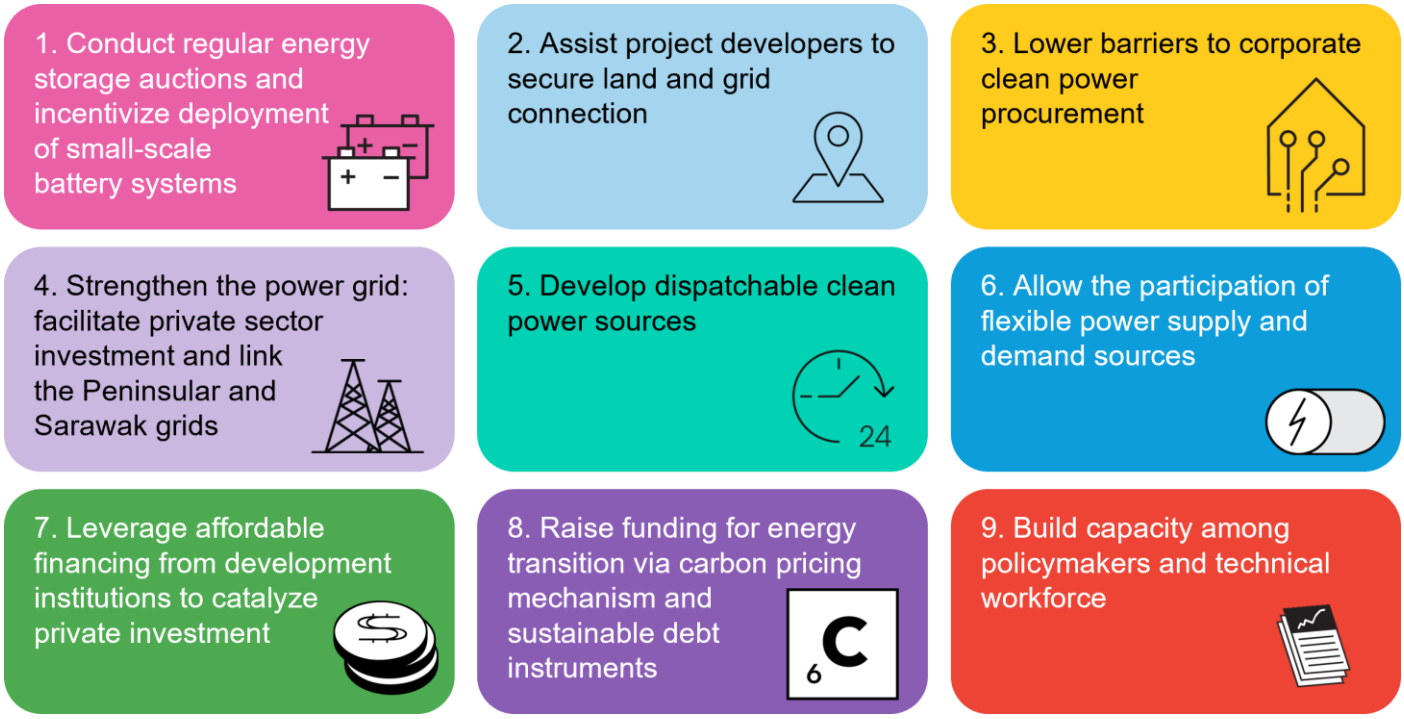


Source: BloombergNEF. Note: Figures for 2024 are estimated. CCS is carbon capture and storage.

5.1. Measures to accelerate renewable power expansion

Malaysia can manage its energy transition and solve the energy trilemma of sustainability, security and affordability by accelerating renewable power additions and grid capacity expansion, while limiting new thermal power capacity addition. In this section, we outline several policy suggestions to further accelerate the growth of clean power in Malaysia (Figure 31).

Figure 31: Suggested measures to accelerate renewables deployment in Malaysia



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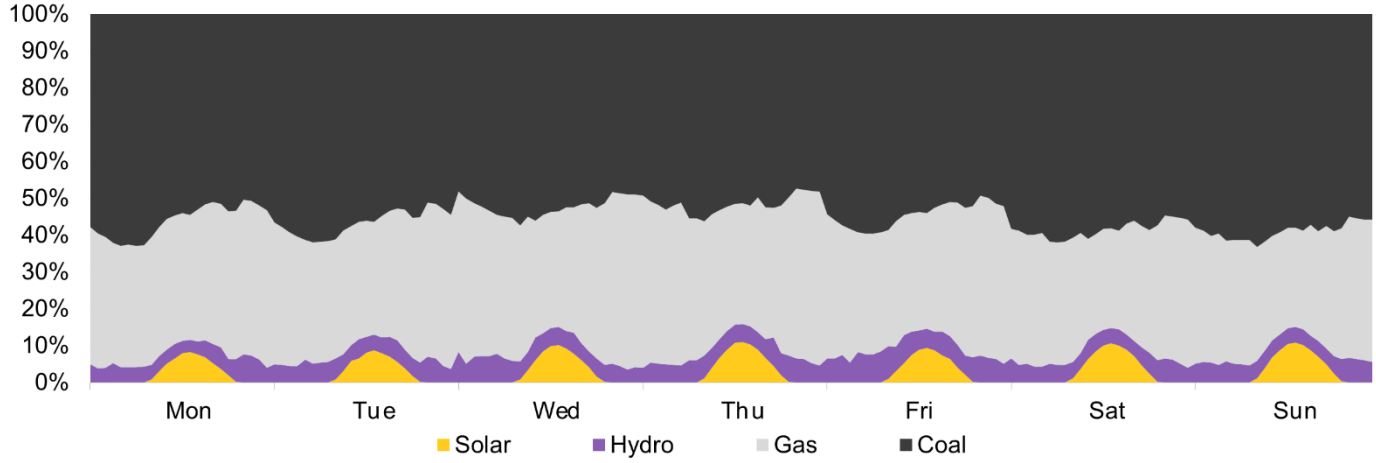
Conduct regular energy storage auctions and incentivize deployment of small-scale battery systems

High levels of solar power should be complemented with energy storage to ensure grid stability and prevent excessive curtailment

Malaysia has signaled its long-term commitment to developing renewable energy through the NETR and NCCP 2.0. Both policy documents place a great emphasis on solar power, which the NETR expects to constitute 58% of the country's power generation capacity by 2050. However, unless complemented by energy storage systems, high levels of solar penetration can pose challenges to grid stability and reliability due to its intermittent nature. Furthermore, a power system that is dominated by solar is likely to experience significant curtailment during the daylight hours, leading to wasted energy and financial loss unless the excess power is stored for later use.

Malaysia has not quite reached this point yet, with solar meeting around 8-11% of power demand at noon in early 2025 (Figure 32). However, the country could ensure the stability of its future power supply system while still having plenty of solar power by encouraging investment into energy storage systems now. This should include both battery energy storage systems (BESS) and pumped hydro storage. BESS can firm up solar power output in the short term (one- to four hour-long storage) and provide ancillary services like frequency and voltage regulation to the grid, while pumped hydro can store excess energy over much longer periods of time.

Figure 32: Hourly power generation mix in Peninsular Malaysia from January 27 to February 2, 2025



Source: BloombergNEF, Single Buyer.

Regular battery storage auctions could boost investor confidence and help build local supply chains

Malaysia's Energy Commission is preparing to hold its first BESS auction later this year, with a goal of commissioning four 100MW/400MWh facilities in 2026. This should be followed by a **clear, long-term pipeline of regular auction rounds for both standalone BESS and solar-with-storage**. This strategy could lower uncertainty and boost investor confidence in the market. Furthermore, regular auctions would also aid capacity planning and could help attract investments to build local supply chains.

In addition, Malaysia could **accelerate the deployment of small-scale standalone batteries and rooftop solar-with-storage** by residences and businesses. This can be done initially through subsidies and rebates. In the longer term, market-based mechanisms such as enabling virtual power plants by allowing aggregation of behind-the-meter resources as well as ensuring electricity consumers are paying the real cost of energy are more financially viable. An initial step could include introduction of time-of-use tariffs. A time-of-use tariff system could incentivize electricity consumers to adjust their electricity consumption pattern with the support of batteries, ultimately leading to a more stable national power demand curve.

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These steps could help Malaysia achieve economies of scale for battery equipment and systems, similar to how the country's Large Scale Solar auctions and Net Energy Metering program have fostered the spread of solar power in the country.

Assist project developers to secure land and grid connection

Governments can de-risk renewable projects by offering suitable land and facilitating grid connections

Land and grid access are two of the largest development risks for renewable project investors. Utility-scale solar projects require large, contiguous and ideally flat land areas. Similarly, hydro and geothermal plants need large and highly specific areas with a suitable resource. Aside from finding parcels of land in areas with viable resources and access to grid interconnection, project developers may also have to compete with agriculture and industrial businesses. In time, as the best sites with the highest-quality renewable energy resources and nearby grid access get used, the challenge of finding suitable sites for new projects increases and can hold back capacity deployment. Even when land is available, project developers could also face acquisition barriers.

The Malaysian government can help to mitigate this barrier by providing a national registry of suitable land and ownership, streamlining permit approval processes, offering available and suitable government-owned land through reverse tenders, and coordinating and facilitating grid infrastructure development and basic civil works for suitable sites, including acquiring required necessary land rights. These steps could help to de-risk projects for independent power producers and spur renewable energy deployment.

Lower barriers to corporate clean power procurement

Malaysia initiated its power market liberalization in October 2024 through the Corporate Renewable Energy Supply Scheme, which allows new commercial and industrial power consumers to procure clean power from private suppliers. This program was then expanded in March 2025 when the government allowed existing power consumers to switch their supplier to a private developer from TNB. However, demand is still relatively limited due to the costly grid system access charges, which are pegged at 0.45 ringgit/kWh ($96/MWh) for non-firm power and 0.25 ringgit/kWh ($54/MWh) for firmed power. Consequently, only two contracts have been signed under this program as of March 2025.

Ease of access to clean power can attract foreign investment into Malaysia

Policymakers should consider lowering the system access charges to boost interest among potential consumers. Malaysia is home to a large base of data centers, manufacturers and suppliers to multinational companies with ambitious sustainability targets. This includes many members of the 24/7 Carbon-Free Coalition and RE100, such as AirTrunk, Samsung and Intel. Providing corporations with affordable clean power procurement options can help Malaysia attract more foreign direct investment by boosting its attractiveness as a technology and manufacturing hub. In turn, growing corporate demand for clean power can help the local solar and especially battery storage industry to achieve greater economies of scale.

Strengthen the power grid: facilitate private sector investment and link the Peninsular and Sarawak grids

A robust and extensive power grid network is a prerequisite for a deep renewables penetration. The planned boom in solar capacity between today and 2050 would require Malaysia's grid to be expanded and strengthened to avoid grid capacity constraints. The responsibility of grid development currently lies solely with TNB, Sarawak Energy and Sabah Electricity, which have a monopoly over power transmission and distribution in their respective operational areas. However, **power grid development can be accelerated by tapping into private capital** to widen the funding pool. This would require policymakers and regulators to draw up a mechanism

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to compensate grid investors, such as through a toll fee levied on electricity transmitted through privately-owned grid systems.

Allowing private entities to assist in building Malaysia's grid can free up some of TNB and Sarawak Energy's capital for other critical projects, such as **building the proposed undersea transmission line between Sarawak and Peninsular Malaysia**. This line could link Sarawak's large hydropower potential with the densely populated and power-hungry states on the peninsula, potentially allowing its rising power demand to be met by Sarawak's hydropower rather than by thermal power plants.

Develop dispatchable clean power sources

Investment into dispatchable clean power is of increasing importance with the rising number of data centers and companies aiming for 24/7 carbon-free energy

Besides building more solar and BESS, **Malaysia also needs to maximize the use of its dispatchable renewable energy sources, such as hydro, bioenergy and nuclear**. This is of increasing importance as the country hosts a rapidly increasing number of data centers requiring a large and continuous supply of power, and as more multinational companies aim to power their operations with 24/7 carbon-free energy.

In the NETR, Malaysia's 6.2GW of hydro capacity is projected to increase to 10.7GW by 2050. While not a small number, it represents just two-thirds of the country's identified hydropower potential. Other than building new hydropower dams, it is also technically possible to develop pumped hydro energy storage systems using Sarawak's existing facilities, for example by connecting the Bakun and Murum dams.

Malaysia has more than 3GW of bioenergy potential from palm oil mill waste, municipal solid waste and biogas. However, the NETR is targeting less than 1GW of bioenergy capacity by 2050. This gap is an opportunity to further reduce power emissions in the country. While biomass plants have been suffering from low capacity factors and high LCOEs, these problems can be ameliorated by building larger plants. Such facilities will have a lower capex on a per-megawatt basis and will be able to negotiate for and aggregate feedstock from multiple palm mills, resulting in a more affordable and reliable biomass supply.

Nuclear power deployment would need long-term strong support from successive governments

Nuclear power does not feature in the NETR, which was published in 2023. However, the government officially recognized nuclear power as a potential energy source in late 2024, with a view of getting the first plant online after 2035. As the preparation, development and construction of nuclear power plants takes more than a decade, strong, long-term support from successive governments and political parties would be needed. BNEF's Net Zero Scenario for Malaysia includes 17GW of nuclear power by 2050. However, that scenario does not consider expansion of interconnections with neighboring economies beyond current capacity. A high-capacity regional grid could reduce the need for nuclear power by connecting locations with an abundant supply of renewables to centers of power demand.

Construction of large hydro dams can have negative environmental and social impacts due to deforestation, loss of biodiversity and forced relocation of local inhabitants, who are often members of indigenous communities. Similarly, building nuclear power plants could trigger local resistance due to concerns of radiation hazard and radioactive waste management. Policymakers, regulators and investors should exercise the utmost care in planning and building hydro and nuclear power projects, such that the people's welfare and the environment are not sacrificed in the pursuit of an energy transition.

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Allow the participation of flexible power supply and demand sources

Energy storage and flexible demand assets can enhance grid flexibility and ease curtailment

The federal and state governments, regulators and power system operators need to prioritize flexibility – from both the supply and demand side – to enable higher renewable generation shares. **Implementing frameworks for the participation of flexible assets in Malaysia's power system** can help the country leverage economical dispatchable generation sources. It can also ease curtailment challenges and enhance grid flexibility for the integration of variable renewable energy sources. This includes the use of energy storage systems and controllable load assets such as virtual power plants, smart electric vehicle chargers and demand response, which can increase the stability of the grid and better align the demand profile with the generation profile of renewable energy plants in Malaysia.

Introducing market-based mechanisms for procurement of ancillary services will be critical to support the uptake of these technologies. Malaysia can learn from power market designs used by other markets such as Australia and Singapore.

Leverage affordable financing from development institutions to catalyze private investment

Low-cost capital can help less mature low-carbon technologies to thrive

Low-carbon technologies tend to be associated with high capital expenditure. The relatively mature ones like solar have enjoyed a larger pool of investment capital as they increasingly come to resemble traditional infrastructure investments rather than risky alternatives. However, technologies such as energy storage and wind power that are less familiar to Malaysian developers and funders may see higher barriers to financing due to the uncertainties around performance, supply chains, risks and profitability. These technologies may require affordable financing and other forms of support from governments and multilateral institutions at the start.

Commercial banks may be guided to invest into certain sectors by the Malaysian government and central bank. To help banks manage some of the risks, **government or development finance institutions can consider spearheading a risk-sharing facility or providing some form of guarantee for the loans**. Development institutions can also help overcome higher financing costs for emerging technologies with concessional finance instruments – including loans, grants, and guarantees offered below market rate – to both decrease the cost of capital and improve access with longer repayment times and lower interest rates for renewables projects. When blended with commercial loans, emerging technology projects could find it easier to obtain financing deals with more palatable and financially viable costs.

Partnerships between domestic and foreign banks for credit lines or syndicated loans can also ease access to capital. This is because the development of some emerging technologies may have reached a more advanced stage in certain markets. For example, the high carbon price in Europe has generated many more hydrogen projects and offtake agreements there compared to in emerging markets like Malaysia. Over time, European banks are likely to become comfortable with financing such deals faster compared to their emerging market peers.

Raise funding for energy transition via carbon pricing mechanism and sustainable debt instruments

In its 2025 budget, Malaysia announced a plan to impose a carbon tax on the iron, steel and energy industries starting from 2026, with a possibility to include the cement, oil and gas industries as well. The proceeds of this tax are earmarked to fund the deployment of low-carbon technologies, including renewables, but the tax rate remains under discussion. Provided it is not set too low, **a robust carbon pricing instrument can be effective in driving renewables installation** and providing a revenue source to fund the energy transition.

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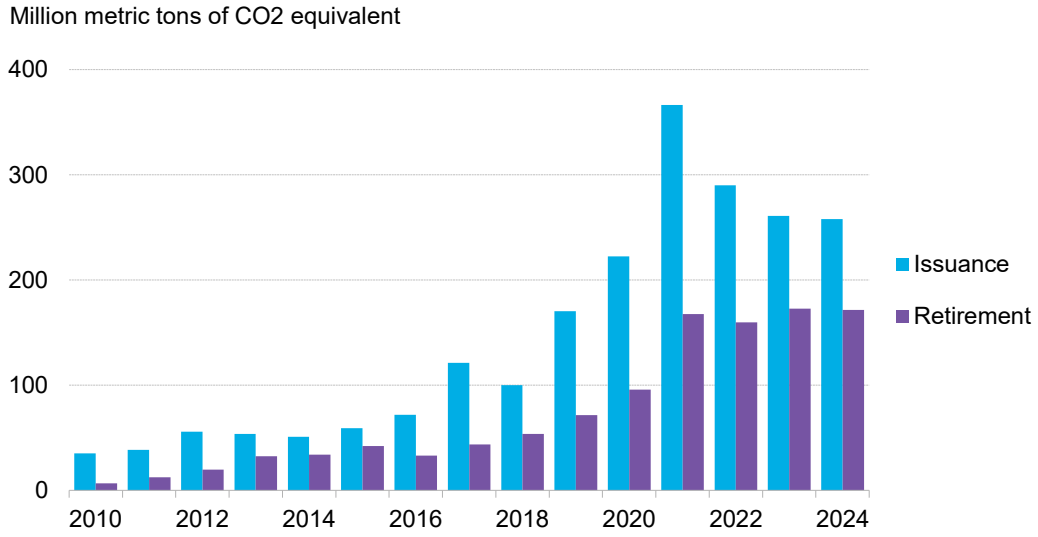
Proceeds from a carbon tax and sales of carbon credits can boost the financial viability of renewables projects

Besides carbon tax, voluntary carbon markets can also help accelerate clean power build-out in Malaysia. The voluntary market, where verified emission reduction credits are traded, has seen interest and activity rise significantly over the past decade, even though credit issuance and redemption volumes have stabilized in more recent years. In 2024, 258 million credits – each equivalent to a metric ton of carbon dioxide – were issued and 172 million credits were retired (Figure 33). Energy generation credits, issued mostly from clean energy projects, made up nearly a third of this supply and 38% of the demand.

Figure 33: Global carbon credit issuance and retirement volumes



Source: BloombergNEF, Verra, Gold Standard, American Carbon Registry, Climate Action Reserve.

Sales of voluntary carbon credits can improve the financial viability of marginally bankable or relatively novel renewables projects in Malaysia, such as solar-with-storage or wind power plants. However, the voluntary carbon markets have come under sharp scrutiny around greenwashing, quality of supply, and lack of transparency in recent years. Ensuring the integrity of Malaysian carbon credits will be critical to ensure that there is a healthy level of demand for them.

Malaysia can also **utilize issuance of sustainable debt instruments** such as green bonds and transition bonds to lower the cost of financing its energy transition. Such instruments can be issued by the government, as Japan had done with its Green Transformation bonds (referred to as GX bonds), as well as by Malaysia's energy companies, such as TNB and Petronas.

Build capacity among policymakers and technical workforce

As renewables and other clean technologies develop rapidly, Malaysia needs to **run capacity building programs** to ensure that government officials and power sector stakeholders have a good understanding of clean power technologies and business models. This would enable them to craft better regulations and a policy landscape that facilitates faster renewables adoption.

Furthermore, renewable energy projects can create plenty of long-term job opportunities. BNEF analysis suggests renewables have a significantly higher employment factor for local labor than thermal power plants. To take full advantage of renewables' job creation dividend, the government will need to ensure relevant training programs are available. It would also need to scale the programs to expand in tandem with the growth of its renewable energy market.

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5.2. Malaysia would benefit from limiting thermal power expansion

Malaysia has committed not to build any new coal plants and to retire all existing plants by 2044. However, the NETR targets an expansion of the country's 16GW gas power fleet to a peak of 30GW by 2045, before declining slightly to 28GW in 2050. This path is unlikely to be the most economical or environmentally sustainable approach to meet Malaysia's future energy demand.

Periodically evaluate the financial viability of thermal power plant pipeline

Although the cost of gas for power generation in Peninsular Malaysia is capped at 35 Malaysian ringgit ($7.50) per million British thermal units (MMBtu) for the first 800 million standard cubic feet per day (mmscfd), any usage above this limit is charged at a cost close to international market prices. Therefore, TNB is still exposed to the global LNG market volatility to some extent. This risk is passed on to power consumers through TNB's Imbalance Cost Pass-Through mechanism, which levies an extra charge when prices of gas fuel increase.

Malaysia should review its pipeline of thermal power plants regularly to verify their necessity and viability

Given this financial risk and the incompatibility of new thermal power plants with emissions abatement ambitions, Malaysia should **have frequent and timely reviews of the feasibility and economic viability of the country's pipeline of gas power plants**. A regular review process should be put in place to ensure that projects that are no longer viable are canceled, allowing Malaysia to choose the most economical options to meet growing power demand. This will also enable the country to stay on track for its decarbonization commitments and net-zero target.

As much as possible, Malaysia should look to **replace some of its planned coal and gas plants with renewables** instead. BNEF's sensitivity analysis shows that the cost parity between renewables and thermal power plants will only be delayed by a few years even with depressed fuel costs and would not change the long-term dynamics that solar is a more economic option for the power sector. See Appendix E for more details.

Reward ancillary services from new thermal plants

If new thermal power plants cannot be avoided, their ability to provide ancillary services should be maximized

The intermittent nature of solar power could disrupt the grid's frequency and voltage stability, which depend on a balance between power supply and demand. Therefore, in as much as new thermal power plants are necessary, Malaysia should aim to maximize these plants' ability to provide rapid frequency and voltage regulation to the grid in order to enable a deeper renewables penetration. Plant operators should be compensated for providing these services, either through a fixed capacity payment in the power purchase agreement or by establishing a market for them. These mechanisms could incentivize power producers to develop and maintain plants with high ramping rates, low minimum uptime and downtime, and the ability to start electricity production without an external energy supply (black start).

Consider earlier retirement of existing coal plants

Malaysia has already committed to phasing out coal power by 2044, which puts it in a leading position among Southeast Asian nations. However, **moving this deadline forward would cut the country's emissions faster** and improve the world's chances of limiting global warming to 1.5C above pre-industrial temperatures. The coal retirement target should also be made legally binding, such that it is less vulnerable to future changes in government.

To alleviate the economic burden on existing power plant owners, Malaysia could consider tapping concessional funding through the Asian Development Bank's Energy Transition

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Mechanism or other green financing programs. It should also **explore using transition credits**, a type of carbon credit that are generated when thermal power plants are retired early. These credits represent the avoided emissions from those plants and the proceeds from their sales can be used to compensate the power plants' owners.

Early retirement of thermal power plants could be made viable through concessional funding and transition credits

The Monetary Authority of Singapore (MAS) has convened the Transition Credits Coalition (TRACTION) to identify system-wide barriers and solutions for transition credits to be utilized as a credible financing instrument. Similarly, the Coal-to-Clean Credit Initiative (CCCI) is an initiative that aims to pilot transition credits to incentivize a just transition away from coal power plants. To that end, CCCI is developing a carbon methodology under Verra and exploring partnerships with several emerging economies. Separately, Gold Standard has created a concept methodology for using avoidance offsets to phase out coal projects and replace them with clean energy.

BNEF's NetZero Pathfinders

BNEF's *Delivering Net Zero: A Framework for Policymakers* is a handbook for governments to design and implement decarbonization strategies. It aims to serve as a fundamental resource for key stakeholders by outlining the policy actions needed to reach net zero. The public NetZero Pathfinders web platform displays a library of the most effective policies to date. These policies, known as best practices, can be replicated in other markets to accelerate progress to net zero.

To help governments navigate potential policy paths to net zero, the Pathfinders actions are split into four pillars:

1. Accelerate deployment of mature climate solutions
2. Support development of new climate solutions
3. Phase out carbon-intensive activities
4. Create appropriate climate transition governance structures

See https://www.netzeropathfinders.com for more.

5.3. Renewables build-out can boost domestic energy security

Malaysia can reduce its reliance on imported coal by having a renewables-rich power system

Countries around the world are putting greater importance on energy security as the global geopolitical landscape grows increasingly volatile. This is especially true for states with limited domestic supplies of coal, oil and gas, as they are more vulnerable to commodity price swings even in times of peace. Malaysia is the world's fifth-largest gas exporter thanks to its vast reserves, but it is a net coal and oil importer. This leaves the country's power generation sector susceptible to sudden shocks in the global coal market. For instance, Indonesia, which supplies around 60% of Malaysia's coal imports, imposed a month-long export ban in January 2022 when coal supplies at its domestic power plants were running low. Malaysia's coal imports from Indonesia that month plummeted by 73% from the previous month (Figure 34), forcing TNB to source coal from other exporters.

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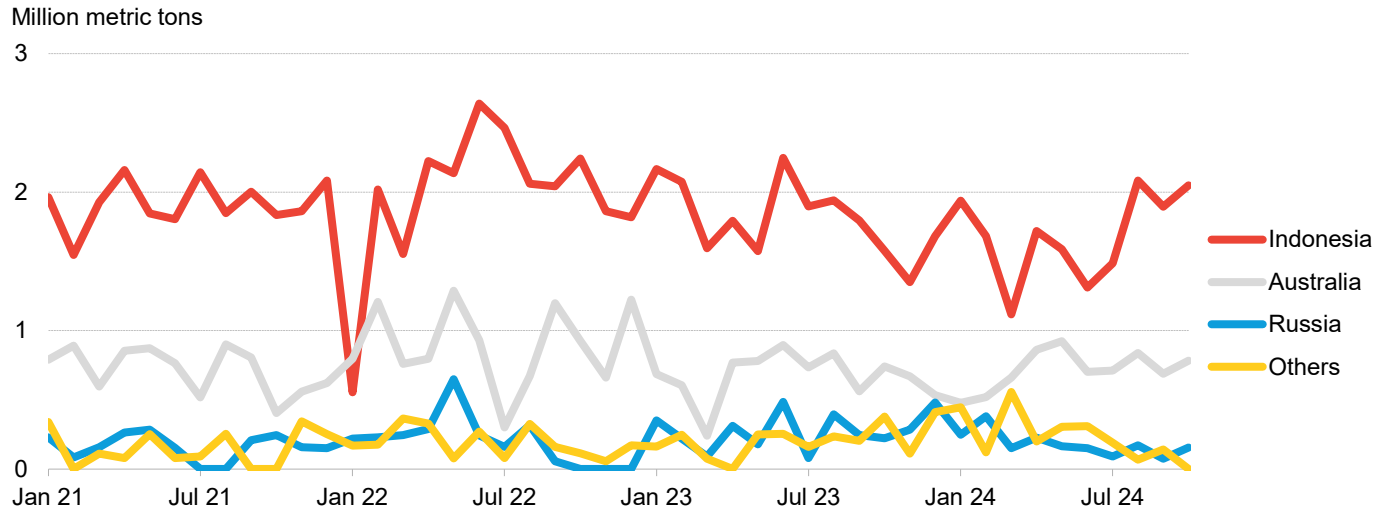
Figure 34: Coal imports to Malaysia, by source



Source: BloombergNEF. Note: 'Others' include Canada, Mozambique, South Africa and the US.

In contrast to coal, oil and gas power plants, renewables like solar, hydro and wind do not require consumable fuels to operate. Therefore, **increasing the role of renewables in the generation mix can reduce Malaysia's reliance on imported fuels and boost its energy security**. Malaysia is also one of the world's top producers of solar modules and inverters, allowing it to expand its solar fleet with little to no imports. Even for hydro and wind power where imported equipment may be needed, the reliance on external supply will be largely limited to the construction phase.

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Appendices

Appendix A. Levelized cost of electricity assumptions

Table 2: Levelized cost of electricity assumptions for benchmark case, nominal values

Technology	Variable	Unit	2025	2030	2050
Coal	Capital expenditure	$ per megawatt (MW)	1,387,491	1,535,178	2,302,753
	Fixed operational expenditure	$/MW/year	39,208	42,554	62,837
	Variable opex	$/megawatt-hour (MWh)	4.5	4.9	7.4
	Capacity factor	%	75.1	76.2	76.2
	Hurdle internal rate of return (IRR)	%	11.6	13.2	14.4
	Cost of debt	Basis points (bps)	668	711	743
	Debt-to-equity ratio	%	68	57	50
	Loan tenor	Years	15	15	15
CCGT	Capex	$/MW	1,018,876	1,127,591	1,691,389
	Fixed opex	$/MW/year	31,314	33,994	50,198
	Variable opex	$/MWh	2.3	2.5	3.8
	Capacity factor	%	56.8	57.7	57.7
	Hurdle IRR	%	10.1	10.3	10.4
	Cost of debt	bps	556	568	576
	Debt-to-equity ratio	%	72	68	65
	Loan tenor	Years	15	15	15
Utility-scale solar	Capex	$/MW	596,536	498,705	493,038
	Fixed opex	$/MW/year	13,191	13,665	17,002
	Variable opex	$/MWh	-	-	-
	Capacity factor	%	16	16	16
	Hurdle IRR	%	9.7	8.5	7.4
	Cost of debt	bps	524	501	493
	Debt-to-equity ratio	%	75	75	75
	Loan tenor	Years	14	14	14

Source: BloombergNEF

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Adjustment for retrofits of fossil fuel power plants

Retrofits of fossil fuel power plants to enable high levels of hydrogen blending and ammonia co-firing require new equipment or facilities added to existing power plants. Table 3 summarizes BNEF's assumptions of adjustments to project costs and efficiency used in our analysis based on interviews with market players and open-source research.

Table 3: Impacts of fossil fuel power plant retrofits for hydrogen blending and ammonia co-firing

	Coal retrofits for ammonia co-firing	CCGT retrofits for hydrogen blending
Capital expenditure (capex)	11% of coal capex	20% of CCGT capex
Variable operational expenditure (opex)	Same as coal variable opex	+20% from CCGT variable opex
Fixed opex	+10% from coal fixed opex	+12.5% from CCGT fixed opex
Efficiency	-6% from coal plant efficiency	-7.5% from CCGT efficiency
Emission reduction	Same as blending ratio of ammonia in energy	Same as blending ratio of hydrogen in energy
Lifetime	20 years	20 years
Financing	Same as a new coal plant	Same as a new CCGT plant

Source: BloombergNEF. Note: Assumes retrofits take place after full depreciation of original power plants. CCGT is combined-cycle gas turbine.

Coal retrofits with **ammonia** include upgrading burners and additional balance of plant expenses to receive and store ammonia. Controlling the exhaust NO_X emissions will also be key to each plant's combustion strategy. Coal retrofits with more than 20% ammonia co-firing have not been tested or commercialized. Hence, our research applied the same retrofit cost assumptions used in Japan's 20% ammonia co-firing as the retrofit costs for more than 20% co-firing, namely 25%, 50%, 75% and 100%. In reality, a higher ammonia co-firing ratio will likely require a higher capex because boilers would require major upgrades or even replacements. Storage tanks for ammonia would also need to be bigger at a higher co-firing ratio. More advanced equipment to capture NO_X emissions would be needed as well. See *Japan's Costly Ammonia Coal Co-Firing Strategy* (web | terminal) for more details.

Hydrogen combustion also requires new equipment including more resilient materials to sustain higher combustion temperatures and more operations and maintenance to deal with these higher temperatures and increased use of water for cooling. The scales of these adjustment-associated costs and efficiency will likely decline over time. For this analysis, we refer to estimated cost and efficiency of a hydrogen-fueled turbine relative to state-of-the-art natural gas turbines between 2019 and 2040. To estimate additional costs and lower efficiency for retrofits, we took simple averages of these two categories and applied the adjustments to CCGT plants. See *Hydrogen: The Economics of Power Generation* (web | terminal) for more details.

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Appendix B. Assumptions for delivered costs of clean fuels relevant to Malaysia

Hydrogen labeling

The hydrogen industry uses labels such as green and blue as a shorthand for how hydrogen is made. Production methods differ on the volume of greenhouse gases they emit. The most common hydrogen labels are:

- Green, made via electrolysis of water using renewable electricity – this process releases few or no greenhouse gas emissions.
- Blue, made via autothermal reforming, steam methane reforming or gasification of coal coupled with carbon capture and storage (CCS) – this releases more emissions than green hydrogen, but less than gray hydrogen.
- Gray, made via autothermal reforming, steam methane reforming or gasification of coal without CCS – the most common method today that releases large volumes of carbon dioxide.

BNEF's research incorporates three different types of clean molecules: green hydrogen/ammonia produced in Malaysia using Sarawak's hydro-dominated grid power, green hydrogen/ammonia imported from Australia and blue hydrogen/ammonia imported from Saudi Arabia. Both Australia and Saudi Arabia are aiming to become clean hydrogen exporters. Australia has cheap solar with vast land areas while Saudi Arabia has substantial oil and gas available as well as potential carbon storage sites. Leveraging on these resources, many companies in these regions have been partnering with hydrogen buyers in other markets to develop hydrogen supply chains.

Our research does not consider fossil fuel-based hydrogen and ammonia without emission mitigation, although almost all ammonia and hydrogen produced today are gray. Use of these molecules without emission abatement defies the justification of promoting these technologies in the first place. Many markets are also encouraging the use of clean molecules by defining 'low-carbon' hydrogen or ammonia.

We estimate fuel hydrogen/ammonia prices by the costs of hydrogen production, ammonia production, conversion back to hydrogen when needed and shipping cost to Malaysia.

Domestic green hydrogen and ammonia production

We used BNEF's *Hydrogen Project Valuation Model* (web | terminal) and *Ammonia Project Valuation Model* (web | terminal) to model domestic green hydrogen and ammonia production in Sarawak, where hydro dams supply about 70% of grid power. Sarawak's electricity supply is not entirely carbon-free, as the remaining 30% is produced from coal and gas combustion. However, the produced molecules are referred to as green hydrogen and ammonia in this report for simplicity. Achieving truly emissions-free hydrogen production would require procurement of renewable electricity through power purchase agreements, renewable energy certificate purchases or other methods. Our cost assumptions are detailed in Table 4.

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Table 4: Domestic green hydrogen and ammonia production assumptions, nominal values

Product	Variable	Unit	2025	2030	2050
Green hydrogen	Capital expenditure	$/MW	1,128,463	996,910	645,360
	Fixed operational expenditure	$/MW/year	28,212	24,923	16,133
	Electrolyzer stack replacement cost*	$/MW	96,999	112,925	254,091
	Electrolyzer utilization rate	%	50	50	50
	Electrolyzer efficiency	Kilowatt-hour per kilogram H2	56	53	47
	Hurdle internal rate of return (IRR)	%	14.3	12.6	10.4
	Cost of debt	Basis points (bps)	820	801	793
	Debt-to-equity ratio	%	62	70	70
	Loan tenor	Years	12	12	12
	Asset life	Years	20	20	20
Green ammonia	Capital expenditure	$/MW	2,938,834	2,703,946	2,211,837
	Fixed operational expenditure	$/MW/year	73,471	67,599	55,296
	Electrolyzer stack replacement cost*	$/MW	138,555	132,903	173,124
	Electrolyzer utilization rate	%	50	50	50
	Electrolyzer efficiency	Megawatt-hour per metric ton of NH3	10.0	9.5	8.4
	Hurdle internal rate of return (IRR)	%	14.3	12.6	10.4
	Cost of debt	bps	820	801	793
	Debt-to-equity ratio	%	62	70	70
	Loan tenor	Years	12	12	12
	Asset life	Years	25	25	25

Source: BloombergNEF

Imported hydrogen and ammonia production

Hydrogen imports are usually transported in the form of ammonia as it is the most economical shipping option.[3] Therefore, we used BNEF's *Ammonia Project Valuation Model* (web | terminal) to derive the production cost of green and blue ammonia to be shipped to Malaysia. Below are the assumptions of technologies used for ammonia production.

[3] Liquid ammonia has a very high hydrogen density (107-121 kilograms of hydrogen per cubic meter) – higher than liquid hydrogen (at 70.8kg per cubic meter). In addition, ammonia can be shipped in a liquid form at -33C, which is technically more manageable than liquid hydrogen that needs to be chilled at -253C.

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- Green ammonia from Australia: alkaline electrolysis using Western electrolyzers powered by a combination of solar, wind and battery storage systems.
- Blue ammonia from Saudi Arabia: autothermal reforming using natural gas, with 95% of emissions abated with a carbon capture and storage system

Transportation of ammonia

Ammonia produced in Australia and Saudi Arabia needs to be shipped to Malaysia. Below is our assumption on transportation costs added to ammonia made in Australia and Saudi Arabia. We used the following estimates for our calculations: 1,653 nautical miles between Dampier in Australia and Malaysia's Pengerang port; and 3,555 nautical miles between Ras Al-Khair in Saudi Arabia and the Sungai Udang port in Malaysia. Ammonia shipping is already mature, so these transportation costs are used throughout the modeling period:

- Molecules from Australia: $21.89 per metric ton of ammonia (in real 2024 dollar terms)
- Molecules from Saudi Arabia: $47.07/ton of ammonia (real 2024)

Conversion of ammonia to hydrogen

Shipped ammonia must be converted back to hydrogen (cracked) if end-use sectors use hydrogen, not ammonia. We estimate that the cracking process has an efficiency of approximately 70.9%, such that 8 metric tons of ammonia are needed to generate 1 ton of hydrogen. Then, we add a fixed cost of $0.6 per kilogram of produced hydrogen to account for the capital and operational expenditures of the cracking process. This estimate is used throughout the modeling period.

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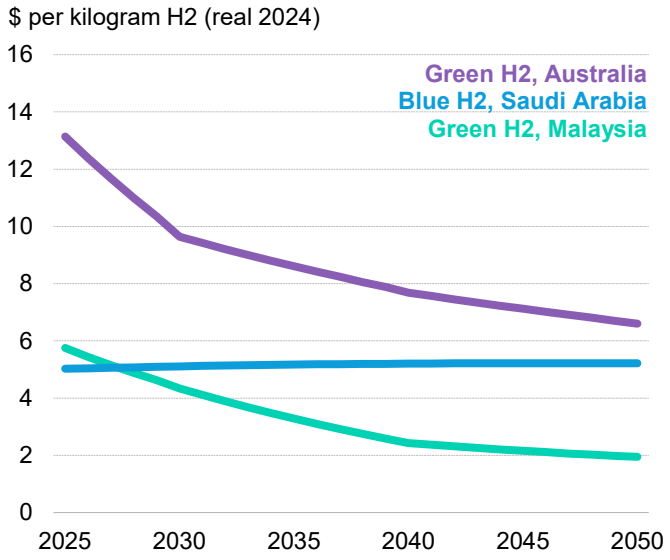
Appendix C. Production cost of hydrogen and ammonia

Green hydrogen produced in Sarawak is the cheapest option from 2028

The levelized cost of green hydrogen produced in Sarawak falls from $7/kg today to just above $2/kg by 2050

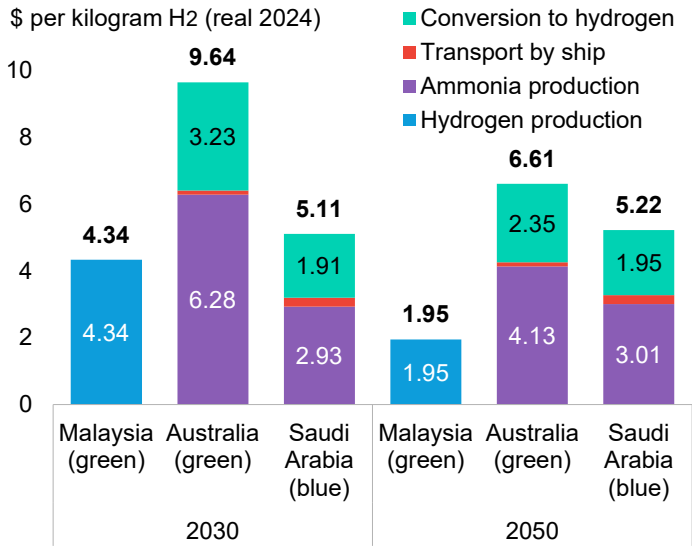
BNEF estimates that green hydrogen produced in Sarawak, Malaysia would cost about $5.8/kg for a project financed this year and just below $2/kg in 2050, supported by Sarawak's very low-cost hydropower (Figure 35). This price range means domestic green hydrogen is already far more affordable than the landed cost of the imported equivalent from Australia and remains so until 2050. While domestic green hydrogen is 14% more expensive than imported blue hydrogen from Saudi Arabia at present, cost parity would be reached by 2028. By 2050, domestic green hydrogen would cost 71% and 63% less than imported low-carbon hydrogen from Australia and Saudi Arabia, respectively, provided the cost of grid power in Sarawak remains broadly stable.

Figure 35: Procurement cost of hydrogen in and delivered to Malaysia



Source: BloombergNEF

Figure 36: Cost of hydrogen supply relevant to Malaysia



Source: BloombergNEF. Note: 'Ammonia production' cost is expressed in $ per kilogram of hydrogen contained in the ammonia molecule.

Ammonia cracking is a costly process that reduces the economic competitiveness of imported hydrogen

One significant cost driver behind imported hydrogen is the ammonia-to-hydrogen cracking process. In our analysis, we assume that hydrogen is exported to Malaysia from Australia or Saudi Arabia in the form of ammonia as it is the most economical shipping option.[4] Once it reaches Malaysia, ammonia is then chemically converted back to hydrogen (and nitrogen). This conversion process is costly, contributing about 30-40% of the landed costs of imported hydrogen (Figure 36).

[4] Liquid ammonia has a very high hydrogen density (107-121 kilograms of hydrogen per cubic meter) – higher than liquid hydrogen (at 70.8kg per cubic meter). In addition, ammonia can be shipped in a liquid form at -33C, which is technically more manageable than liquid hydrogen that needs to be chilled at -253C.

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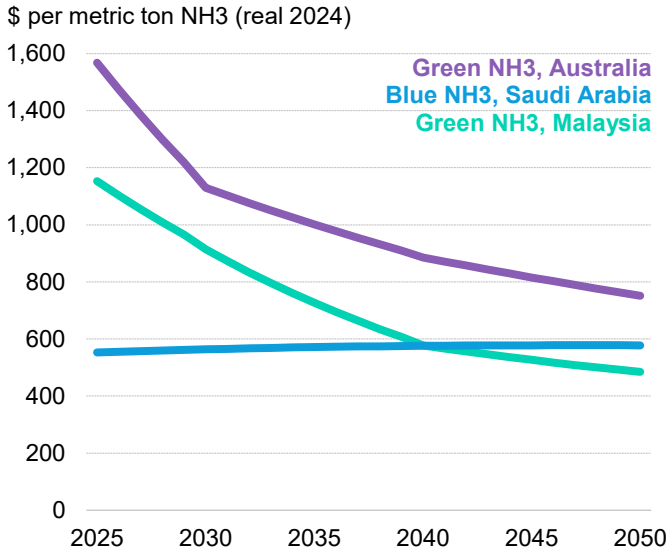
We have not considered the scenario of domestically produced blue hydrogen in Malaysia. Directly using CCS to capture emissions from thermal power plants would be cheaper than using CCS to capture emissions from the process of converting natural gas to hydrogen or ammonia, and then using that resulting blue hydrogen/ammonia in thermal power plants. Applying CCS directly to the thermal power plant is a more energy efficient process than using CCS for hydrogen production and then running thermal power plants on hydrogen or its derivative.

Domestic green ammonia undercuts blue ammonia imported from Saudi Arabia in 2041

Imported blue ammonia is the cheapest option today, but local green ammonia becomes cost-competitive in the early 2040s

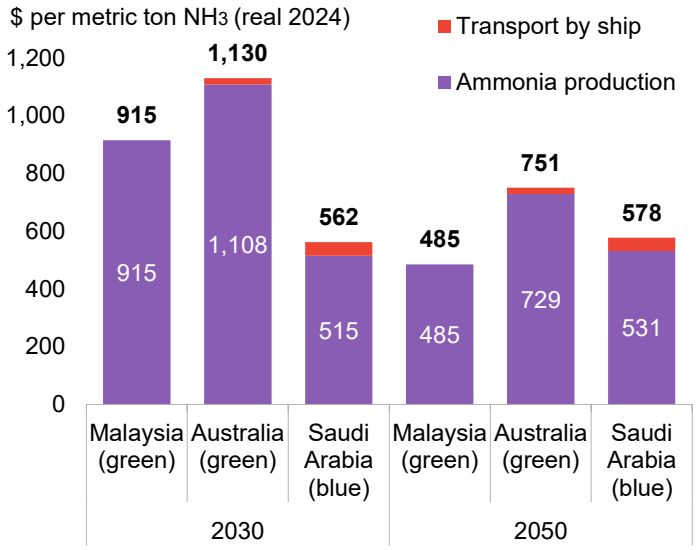
Imported blue ammonia from projects financed this year in Saudi Arabia would cost less than half of green ammonia produced in Sarawak or imported from Australia, due to the low cost of natural gas there (Figure 37). Between the two green ammonia options considered here, domestic production is more affordable than importing from Australia throughout the modeled period. (Figure 37 and Figure 38). Production cost of local green ammonia in Malaysia falls over time, and local ammonia production plants financed in 2041 would undercut the gradually rising cost of blue ammonia from Saudi Arabia. Blue ammonia (as well as blue hydrogen) has limited cost reduction potential because of constrained cost reductions of fossil fuels in the future.

Figure 37: Procurement cost of ammonia in and delivered to Malaysia



Source: BloombergNEF. Note: NH3 is ammonia.

Figure 38: Cost of ammonia supply relevant to Malaysia



Source: BloombergNEF. Note: NH3 is ammonia.

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Appendix D. Blended and co-fired fuel prices

Hydrogen – natural gas blended fuel prices, by blending ratio

Figure 39: Blended fuel price for 25% hydrogen and 75% natural gas mix

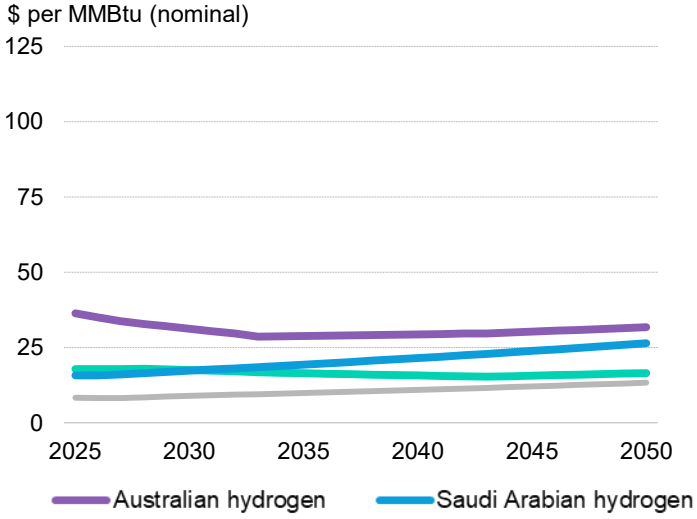
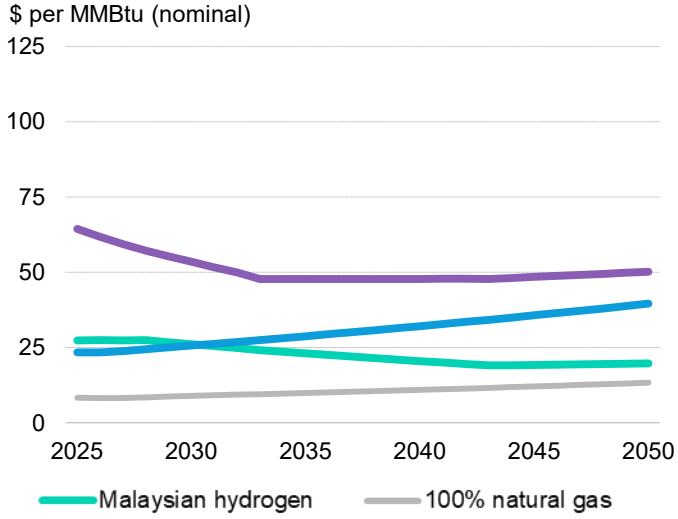


Figure 40: Blended fuel price for 50% hydrogen and 50% natural gas mix



Source: BloombergNEF. Note: Blending ratio is based on energy content. MMBtu is million British thermal units.

Figure 41: Blended fuel price for 75% hydrogen and 25% natural gas mix

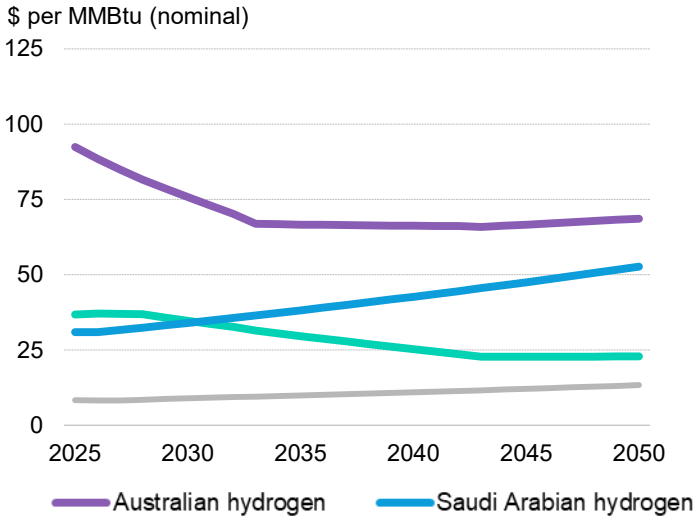
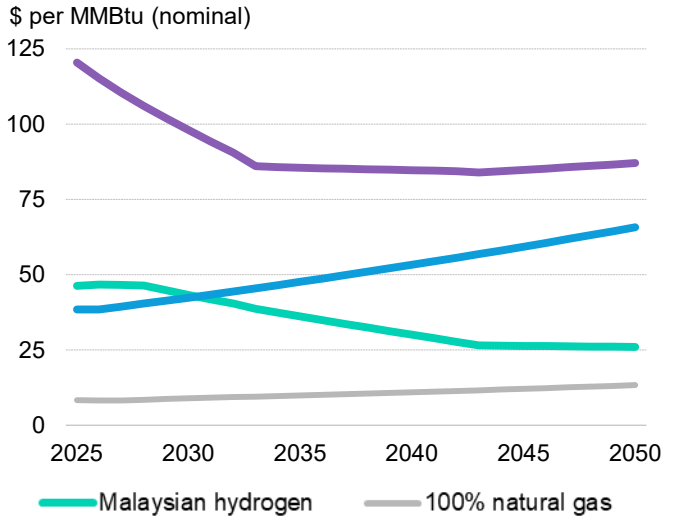


Figure 42: Fuel price for 100% hydrogen



Source: BloombergNEF. Note: Blending ratio is based on energy content. MMBtu is million British thermal units.

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Ammonia – coal co-fired fuel prices, by co-firing ratio

Figure 43: Blended fuel price for 25% ammonia and 75% coal mix

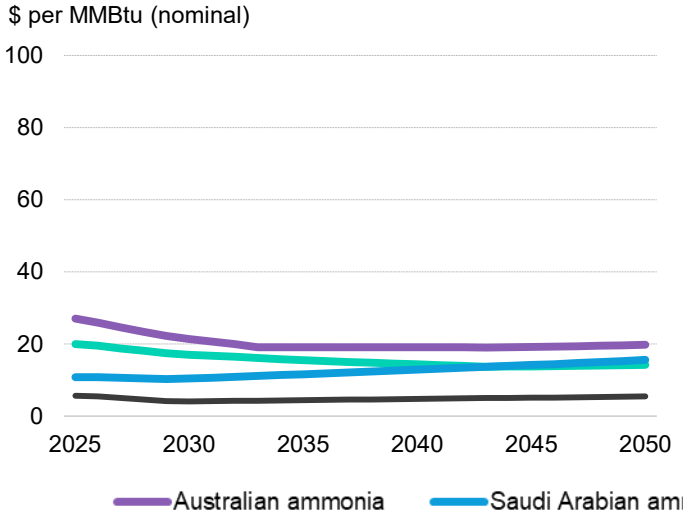
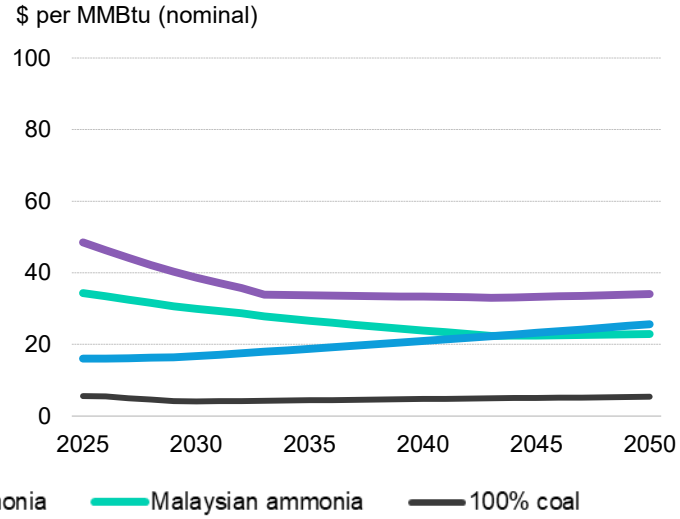


Figure 44: Blended fuel price for 50% ammonia and 50% coal mix



Source: BloombergNEF. Note: Co-firing ratio is based on energy content. MMBtu is million British thermal units.

Figure 45: Blended fuel price for 75% ammonia and 25% coal mix

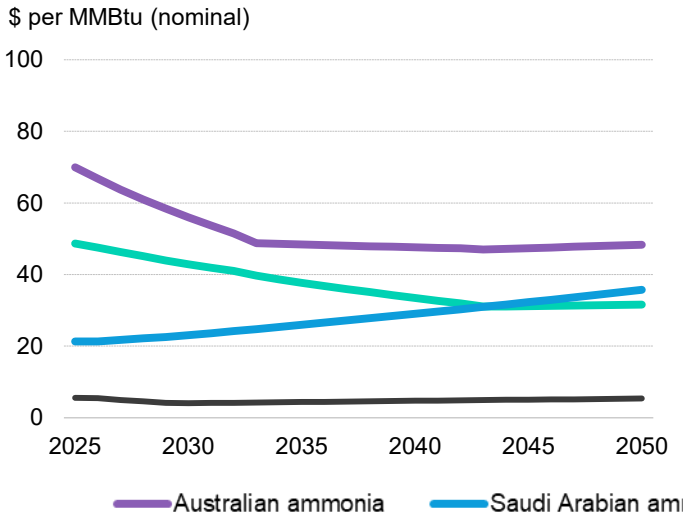
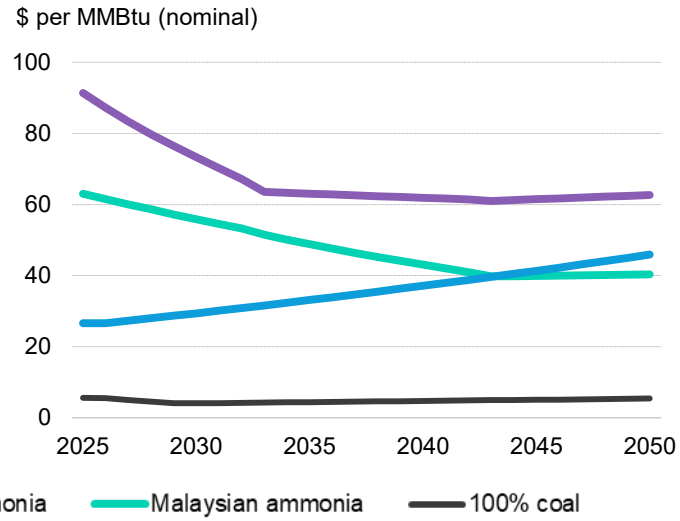


Figure 46: Fuel price for 100% ammonia



Source: BloombergNEF. Note: Co-firing ratio is based on energy content. MMBtu is million British thermal units.

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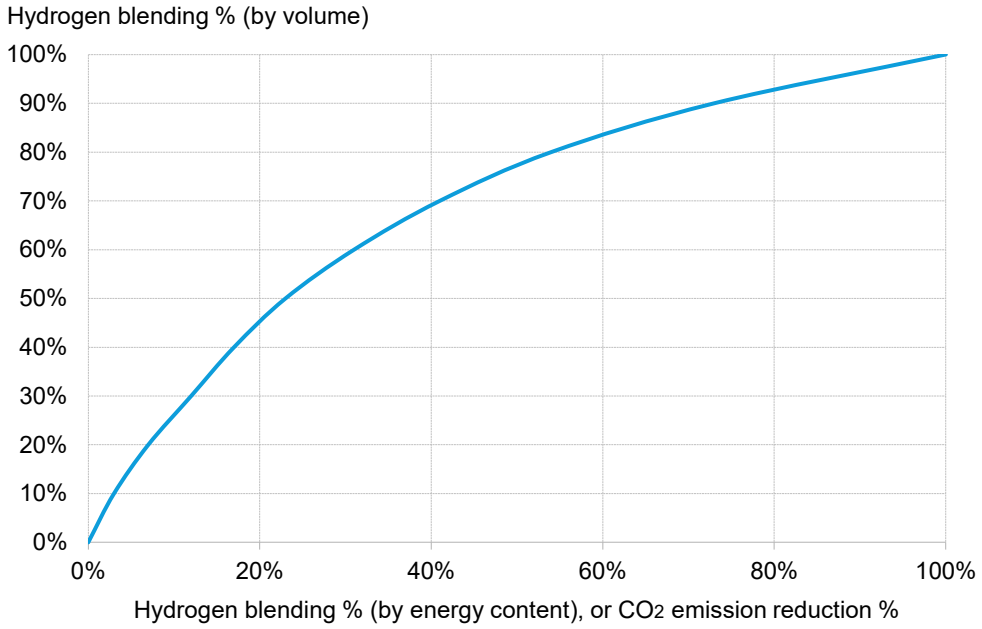
Energy density of hydrogen

As hydrogen has a lower volumetric energy density than natural gas, higher volumes of hydrogen than natural gas are required to achieve a similar energy blend ratio. As a result, to significantly reduce CCGT carbon dioxide emissions, higher volumes of hydrogen than natural gas would have to be consumed. Throughout this report, we use a blending ratio based on energy content.

Figure 47: Relationship between energy ratio and volume ratio for hydrogen blending



Source: BloombergNEF, GE Power to Gas: Hydrogen for Power Generation.

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Appendix E. Sensitivity analyses

Depressed fuel costs are insufficient for thermal plants to compete with the cost evolution of a new solar plant

As the growth of cost-competitive renewables displaces coal and gas power generation, it is possible that less global demand could cut coal and gas prices, resulting in lower LCOEs and marginal running costs of the fossil fuel power plants. On the other hand, geopolitical tensions could raise fuel prices, increasing the LCOE further. To explore how the competitiveness landscape of the different power generation technologies would look like in such a scenario, the LCOE of a new coal and CCGT plant was calculated with the fuel cost set at a 20% premium to a discount of 40% against the benchmark case, which is covered in Section 3.

A 40% drop in seaborne thermal coal prices would lower the LCOE benchmark of a new coal plant by 16.1% on average throughout the forecast period (Figure 48). Even with such an extreme drop in fuel cost, the tipping point where power from a new utility-scale solar plant becomes cheaper than that from a new coal plant will still be reached this year. The same magnitude of coal price decline would push the crossover point between the LCOEs of solar-with-storage and coal plants by only four years to 2032 from 2028.

Compared to coal, a reduction in fuel price has a more significant impact on gas. The LCOE of a new CCGT plant could be reduced by an average of 25.7% against the benchmark case throughout the forecast period with a 40% fuel cost discount (Figure 49). However, even in this case, solar power remains cheaper from today to 2050, while the start of solar-with-storage's cost advantage over gas power is delayed to 2030 from this year.

Figure 48: Levelized cost of electricity of new solar and solar-with-storage plants versus range of LCOEs for a new coal plant in Malaysia

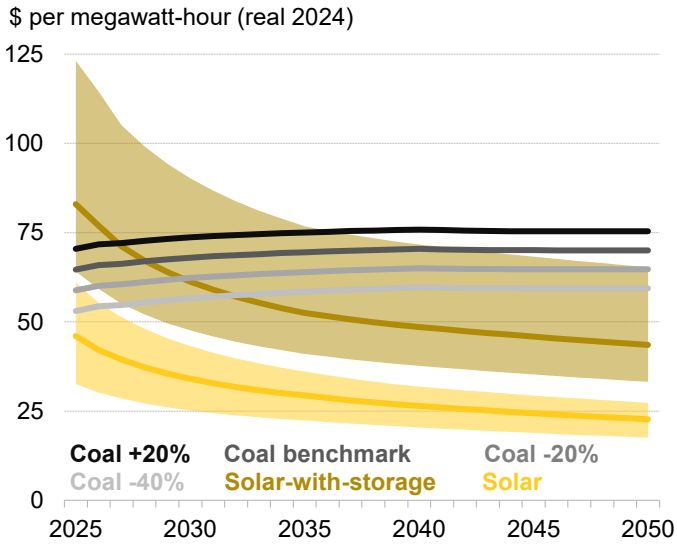
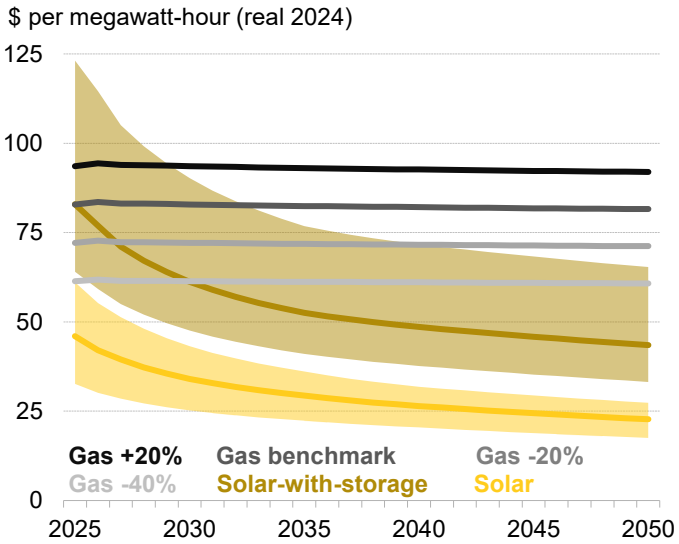


Figure 49: Levelized cost of electricity of new solar and solar-with-storage plants versus range of LCOEs for a new combined-cycle gas turbine in Malaysia



Source: BloombergNEF. Note: Range of coal and combined-cycle gas turbine levelized cost of electricity (LCOE) depicts a fuel cost change of up to a 20% premium and 40% discount from the benchmark fuel cost.

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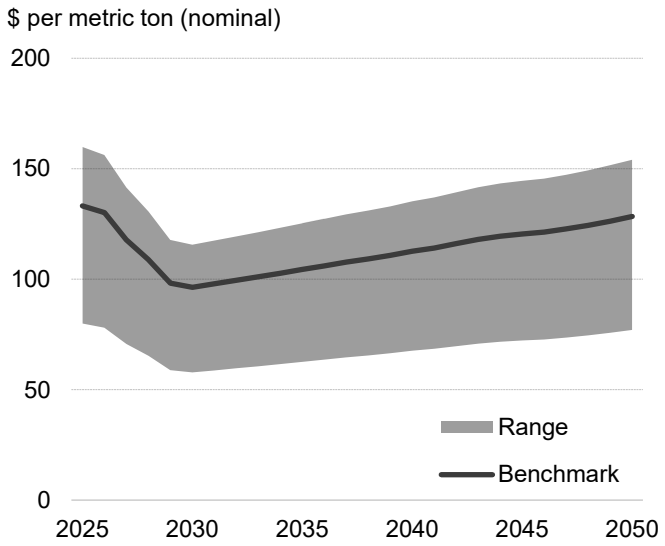
Significant fuel price reduction would be needed to keep existing coal plants competitive

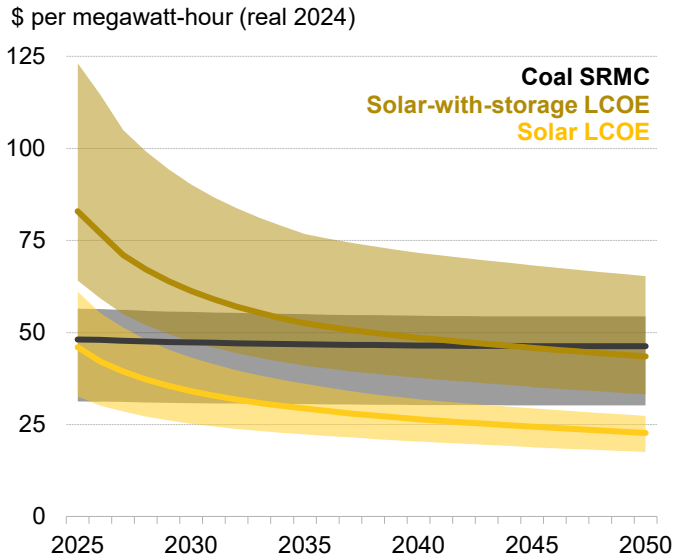
Due to the falling cost of renewables, fossil fuel power plants could be stranded in the future as expensive power generation sources. Without significant fuel price reductions, thermal power plants wouldn't be economically viable in Malaysia's power system. For instance, a 40% drop in coal price against our benchmark fuel price scenario (an average drop of $45.6/ton in nominal terms between 2025 and 2030) could keep existing coal plants economically competitive with new solar-with-storage plants up to 2050 (Figure 51). However, even with such a decline in coal cost, the LCOE of a new solar plant will still be cheaper than the SRMC of an existing coal plant by 2033, and possibly as early as 2026.

Figure 50: Range of coal prices used for sensitivity analysis



Source: BloombergNEF. Note: Range of coal prices represents a 20% premium on the upper end and a 40% discount on the lower end against the benchmark price.

Figure 51: Levelized cost of electricity of new solar and solar-with-storage plants versus short-run marginal cost of an existing coal plant in Malaysia



Source: BloombergNEF. Note: Range of coal short-run marginal cost represents a 20% premium on the upper end and a 40% discount on the lower end against the benchmark fuel price.

The LCOE of a new solar plant in Malaysia is already 30% cheaper than the SRMC of an existing CCGT plant today, while solar-with-storage LCOE will undercut gas SRMC by 2027 (Figure 53). Even with a 40% fuel cost reduction, running existing CCGT plants will be more expensive than building new solar and solar-with-storage plants from 2026 and 2047 onwards, respectively.

Other factors could also negate the effect of fuel price reduction, such as higher-than-expected financing costs for fossil fuel power assets or the 2026 introduction of a carbon price for power generation in Malaysia.

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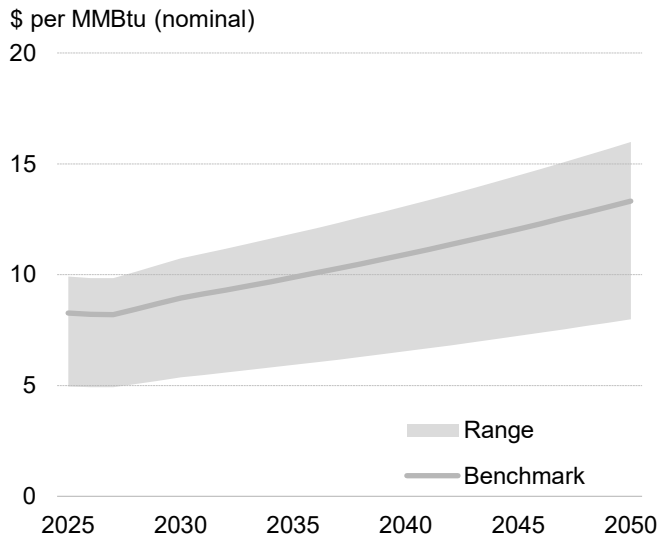
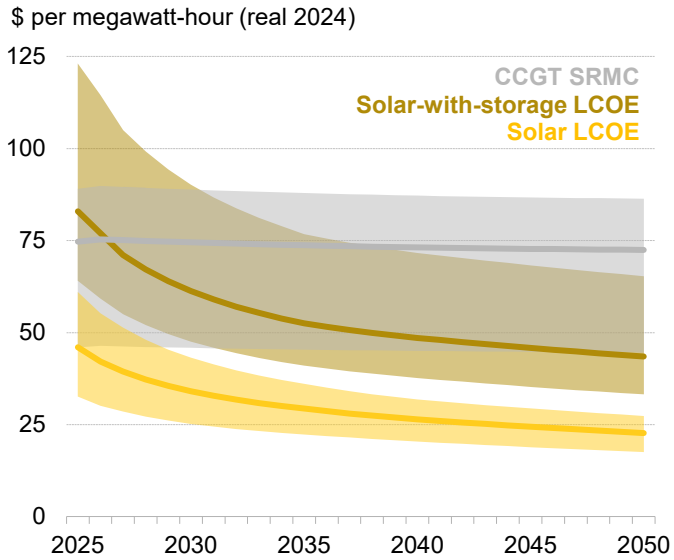
Figure 52: Range of gas prices used for sensitivity analysis



Source: BloombergNEF. Note: MMBtu is million British thermal units. Range of gas prices represents a 20% premium on the upper end and a 40% discount on the lower end against the benchmark price.

Figure 53: Levelized cost of electricity of new solar and solar-with-storage plants versus short-run marginal cost of an existing combined-cycle gas turbine power plant in Malaysia



Source: BloombergNEF. Note: Range of CCGT short-run marginal cost represents a 20% premium on the upper end and a 40% discount on the lower end against the benchmark fuel price.

Changes in capacity factors has a significant impact on the LCOE of thermal plants

When considering the cost competitiveness of different power generating technologies, it is imperative to consider the potential realized capacity factor of each plant due to its significant contribution to the LCOE, especially that of a fossil fuel power plant, instead of the technical potential of the plant.

BNEF's analysis shows that capacity factors have a significant impact on the LCOEs of coal and CCGT plants. A 5% increase in the capacity factor lowers the LCOE of a coal plant by 2.32% while a 5% reduction in capacity factor results in a 2.57% rise in LCOE (Figure 54). For a CCGT plant, a 5% hike in the capacity factor reduces the LCOE by 1.55% while a 5% drop in capacity factor results in a 1.72% rise in LCOE (Figure 55).

The running hours of thermal power plants in Malaysia are already being squeezed by the increased grid penetration of renewables, in particular solar and hydro. In 2022, the capacity factors of the coal and gas power plant fleets in Malaysia were 71% and 48%, compared with 78% and 55% in 2016. This is despite a 21% increase in total electricity generation in 2022 compared with 2016 levels. As Malaysia targets more renewables development, thermal power plants will likely see their operational hours being cut further. This will lead to more costly coal and gas power, as shown in Figure 54 and Figure 55.

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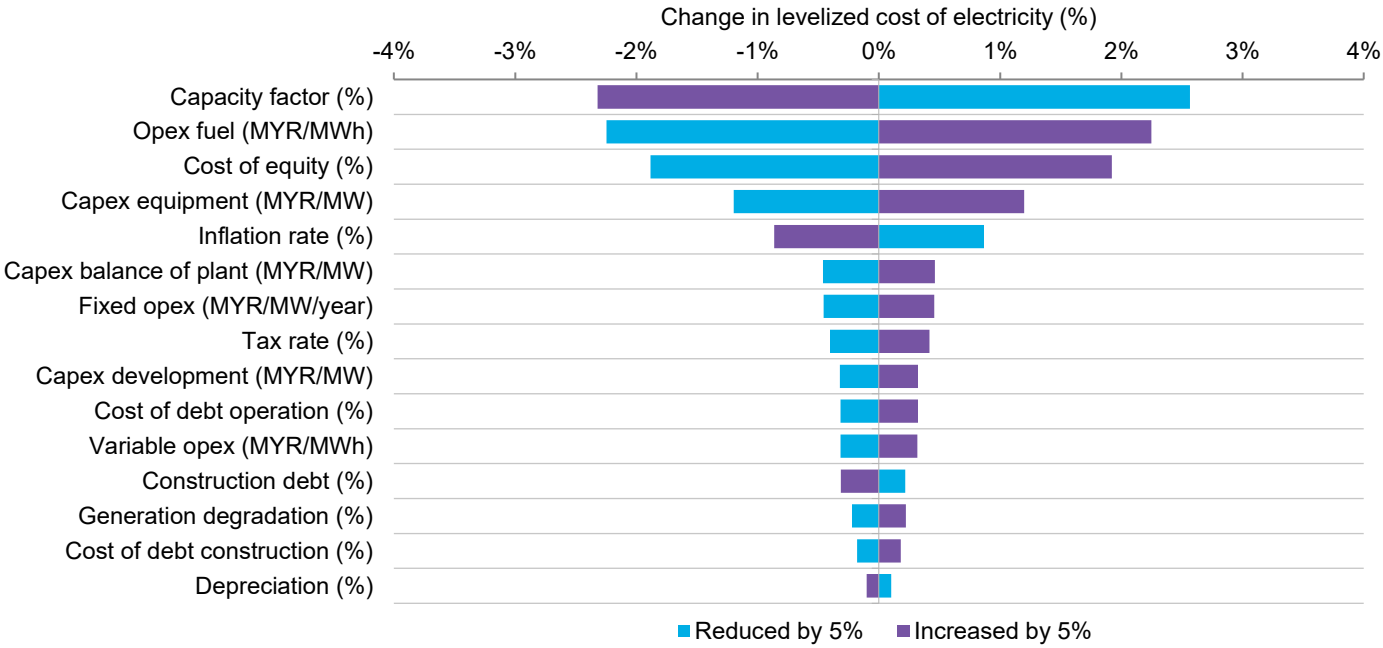
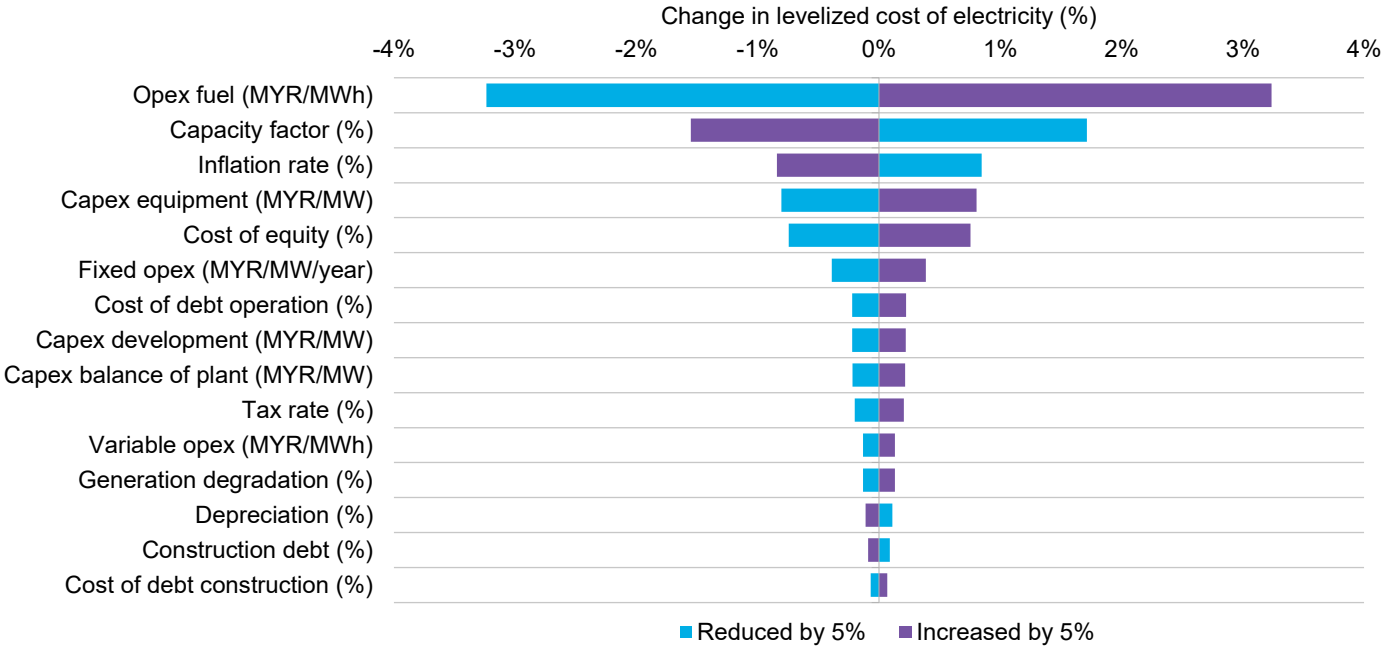
Figure 54: Sensitivity analysis of the levelized cost of electricity of a coal power plant in Malaysia



Source: BloombergNEF. Note: Chart shows the percentage impact on the LCOE of a coal plant in Malaysia with a +/-5% variance on each variable based on the benchmark cost scenario. MYR is Malaysian ringgit, MWh is megawatt-hour, MW is megawatt.

Figure 55: Sensitivity analysis of the levelized cost of electricity of a combined-cycle gas turbine power plant in Malaysia



Source: BloombergNEF. Note: Chart shows the percentage impact on the LCOE of a CCGT plant in Malaysia with a +/-5% variance on each variable based on the benchmark cost scenario. MYR is Malaysian ringgit, MWh is megawatt-hour, MW is megawatt.

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